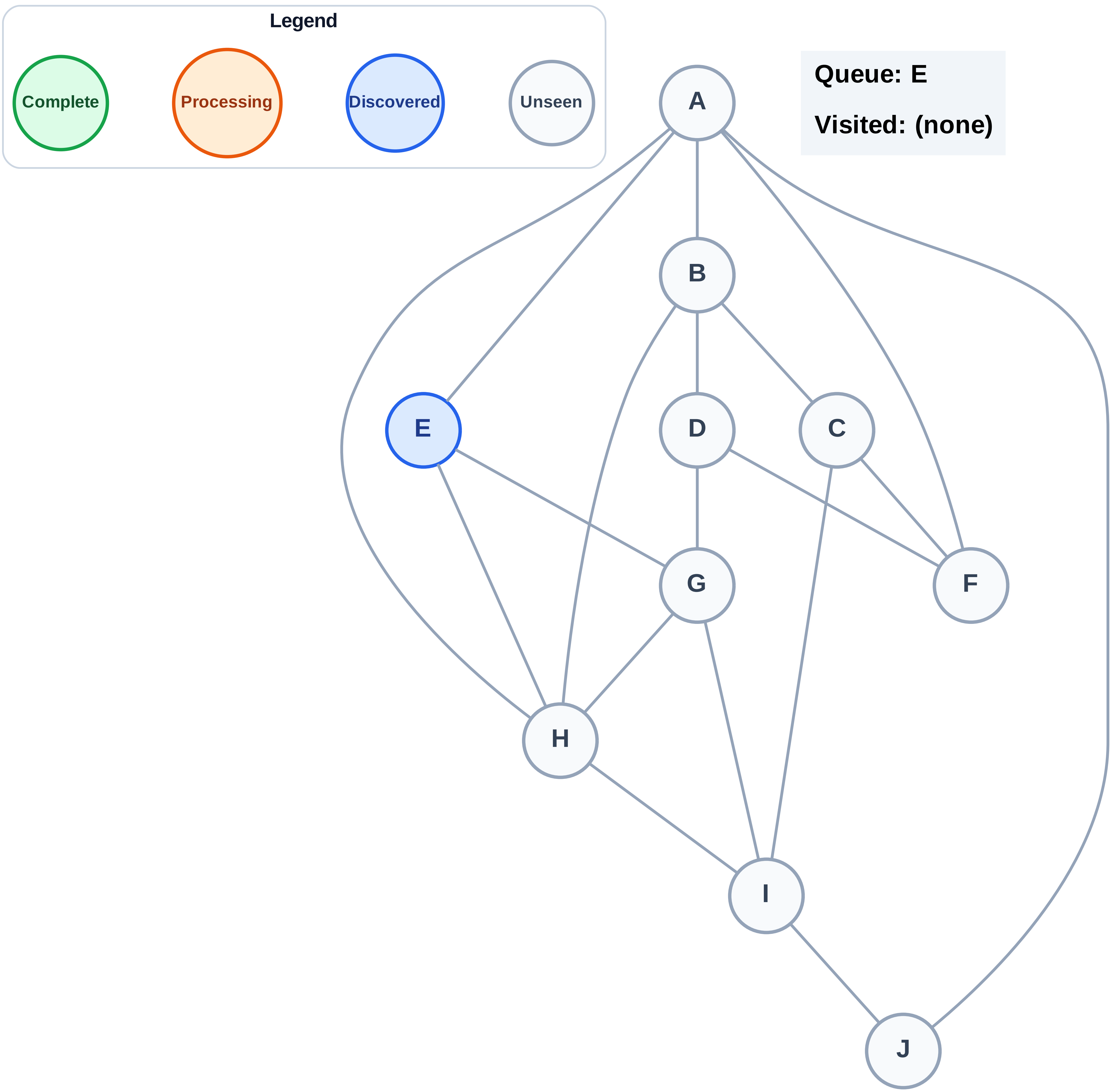
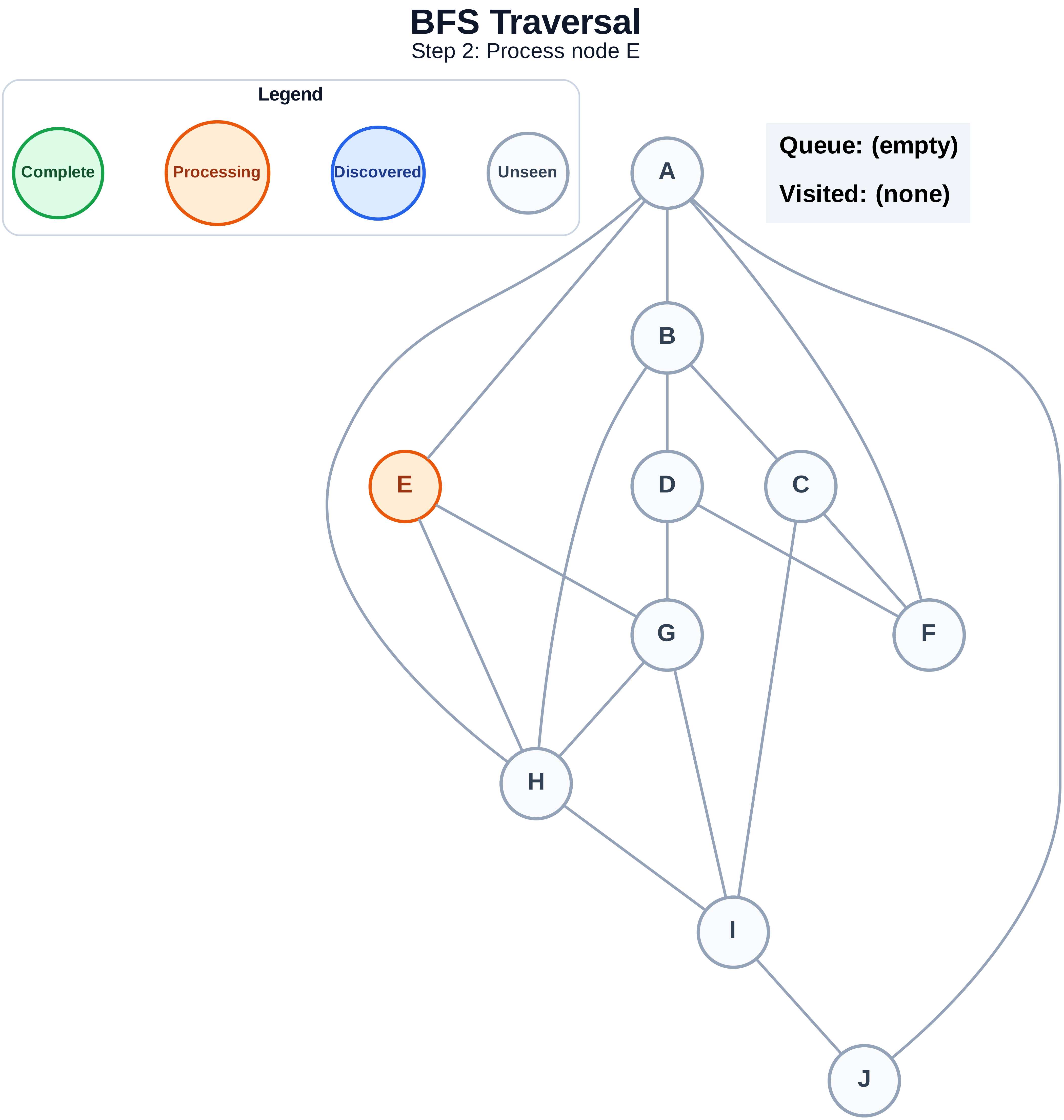
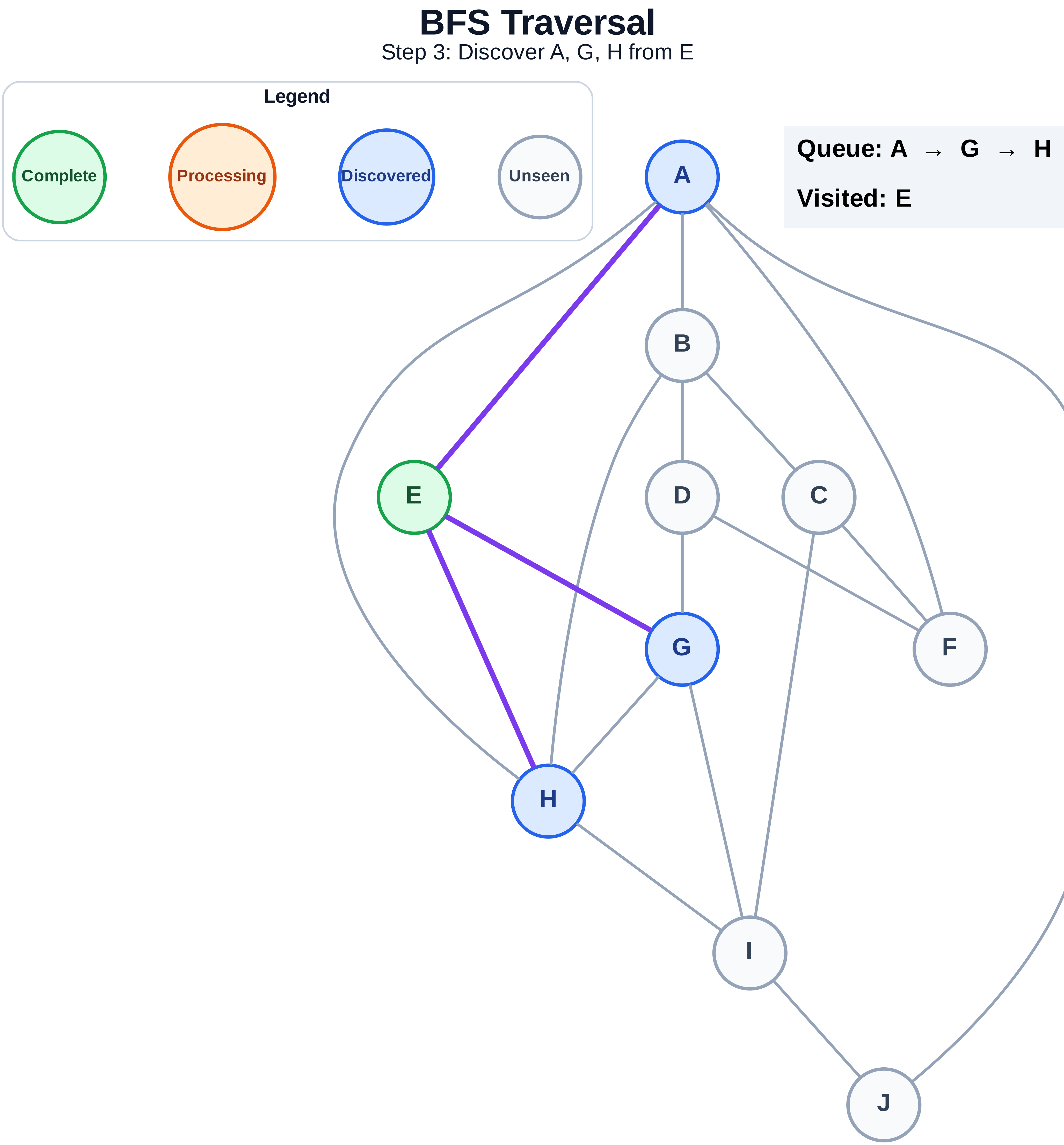
BFS Traversal

Step 1: Start at E



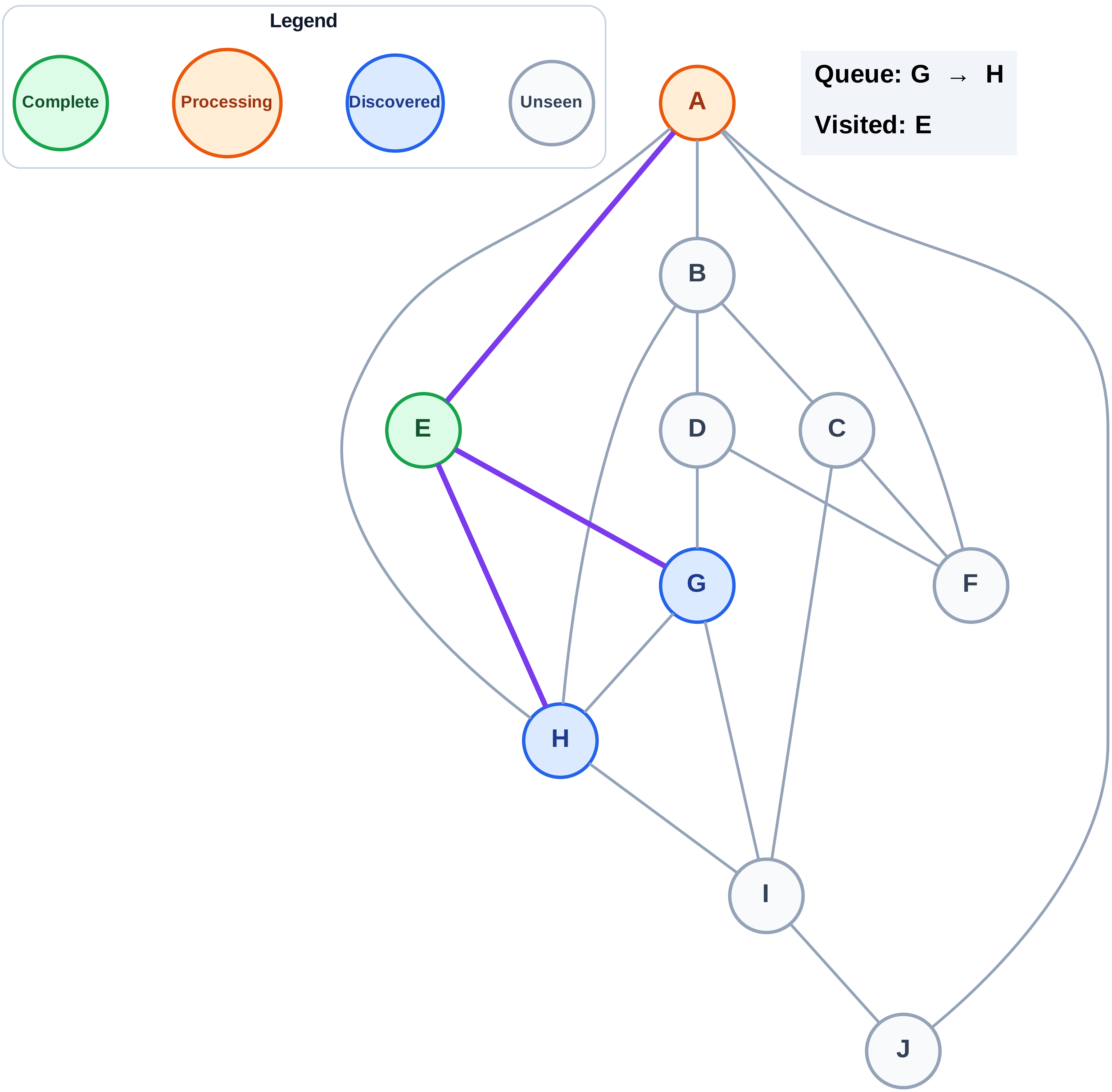

BFS Traversal
Step 2: Process node E
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: (none)
A
B
C
D
E
F
G
H
I
J



BFS Traversal
Step 3: Discover A, G, H from E
Legend
Complete
Processing
Discovered
Unseen
Queue: A → G → H
Visited: E
A
B
C
D
E
F
G
H
I
J

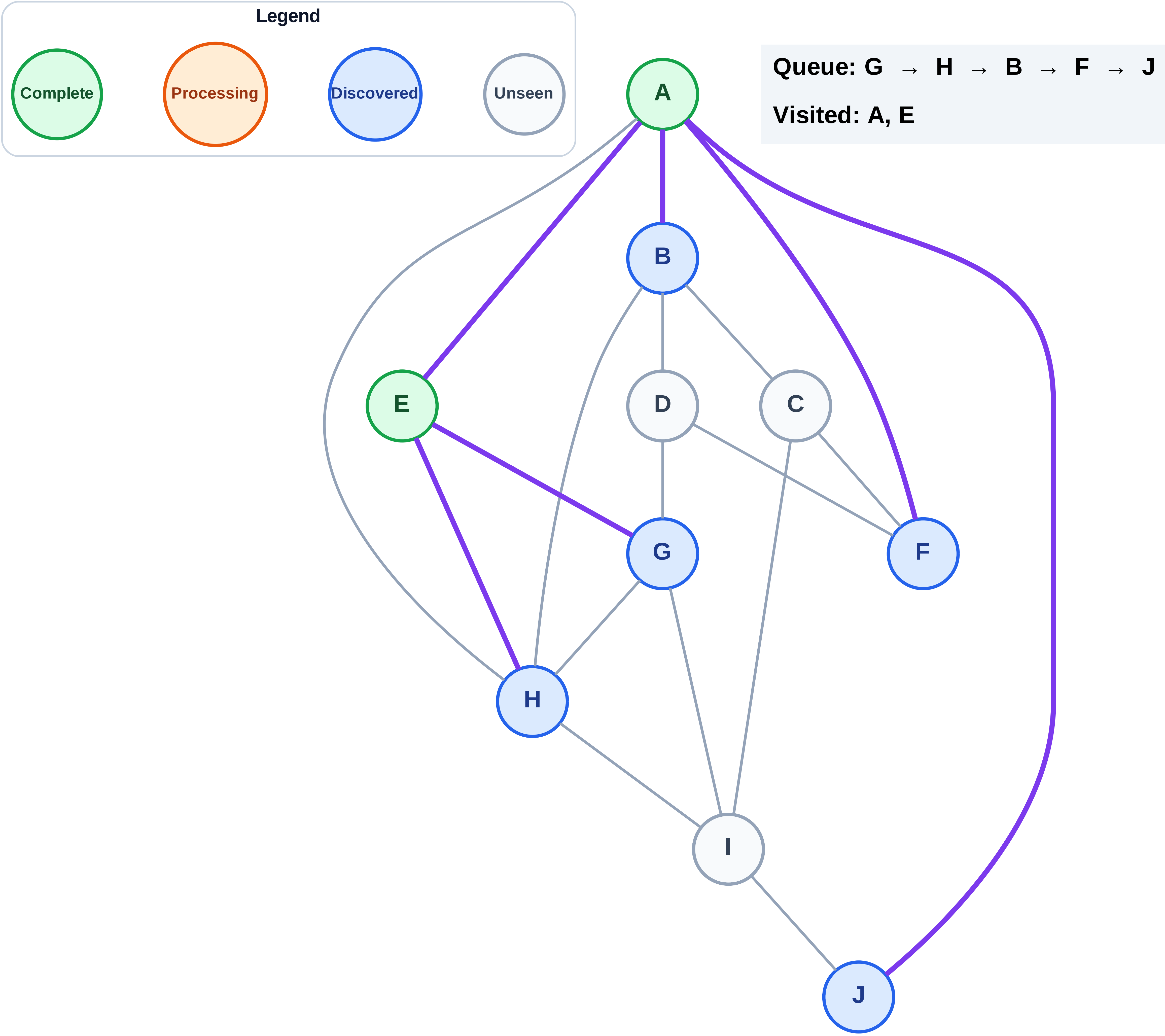
BFS Traversal

Step 4: Process node A



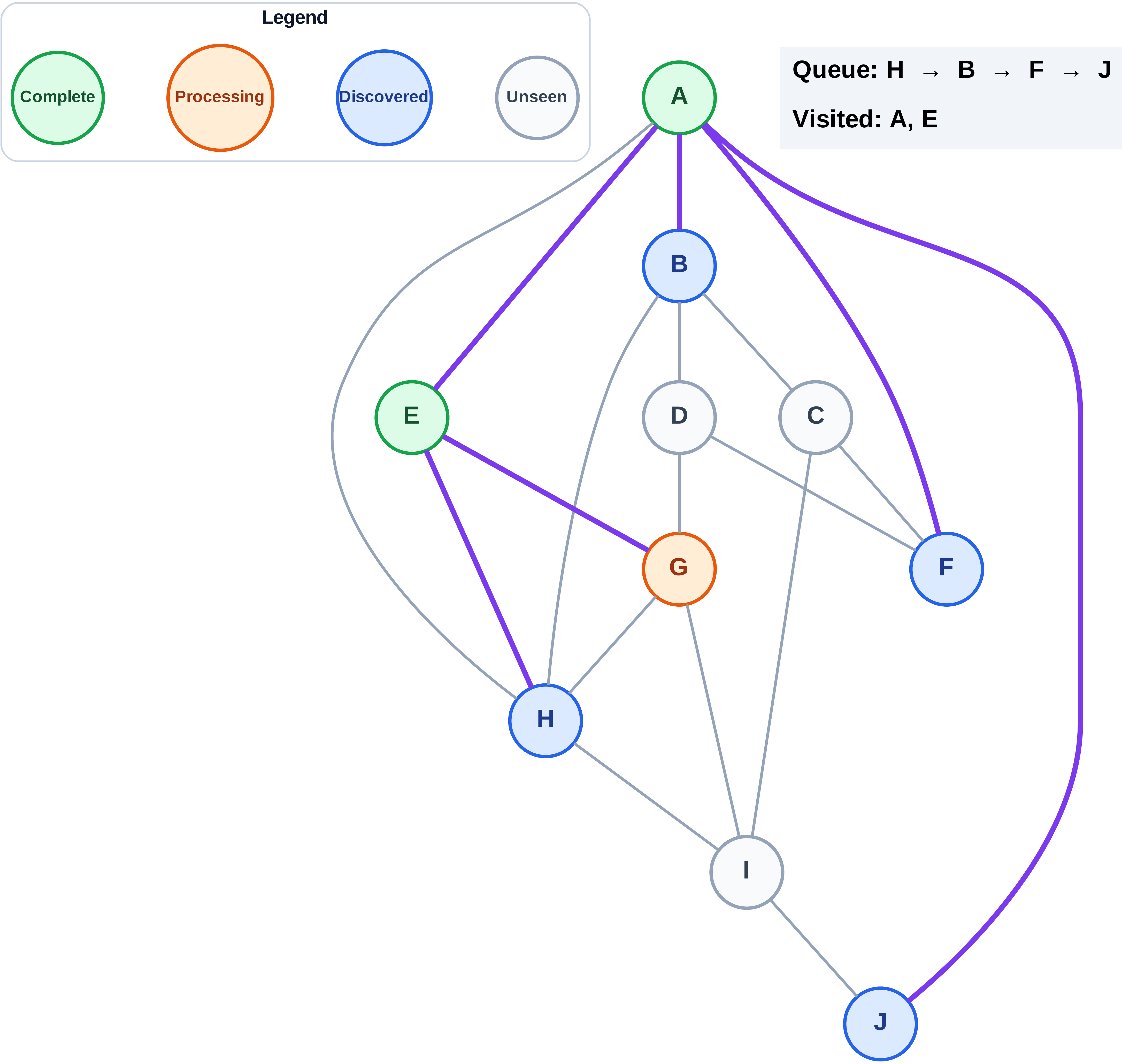
BFS Traversal

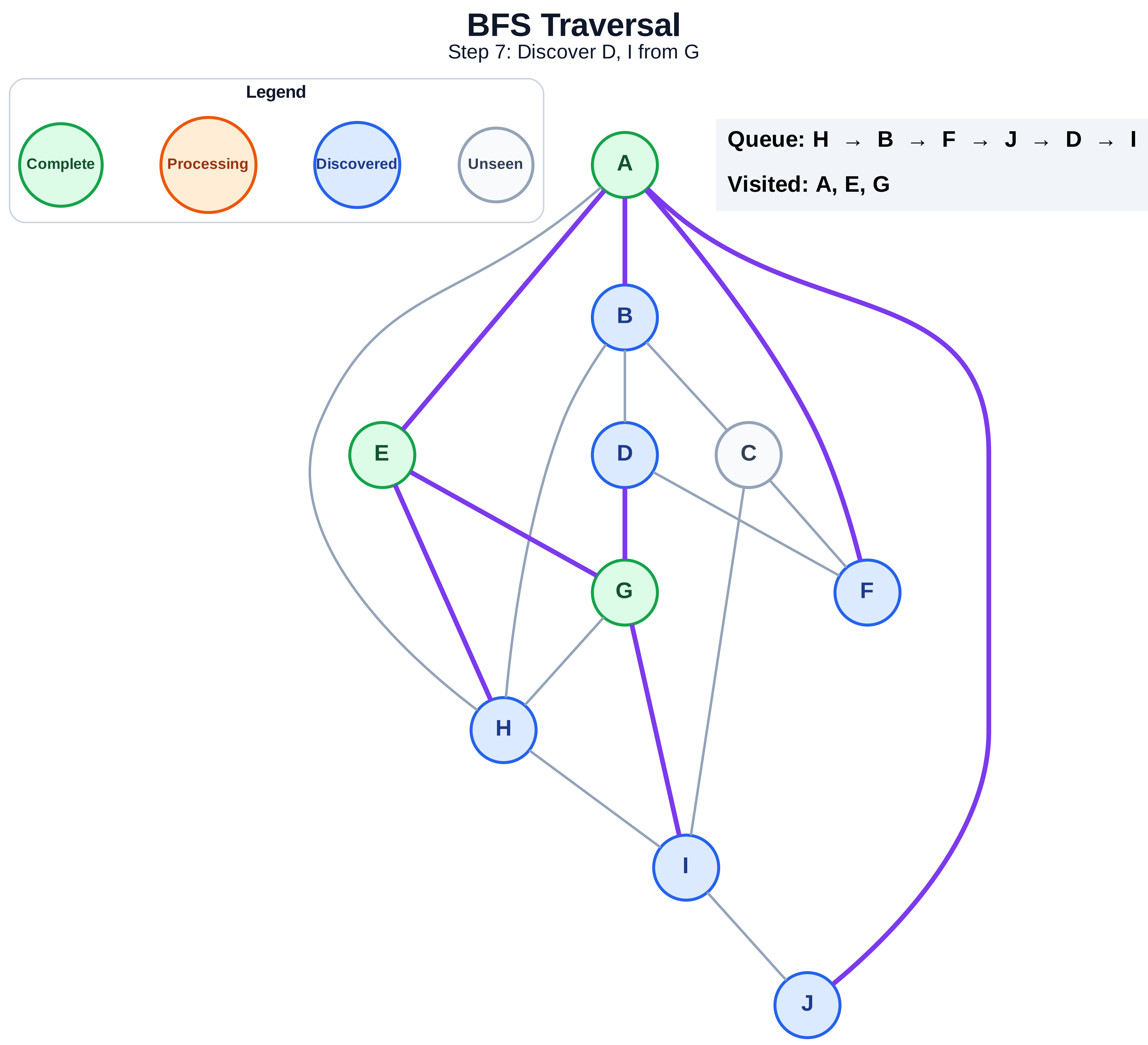
Step 5: Discover B, F, J from A



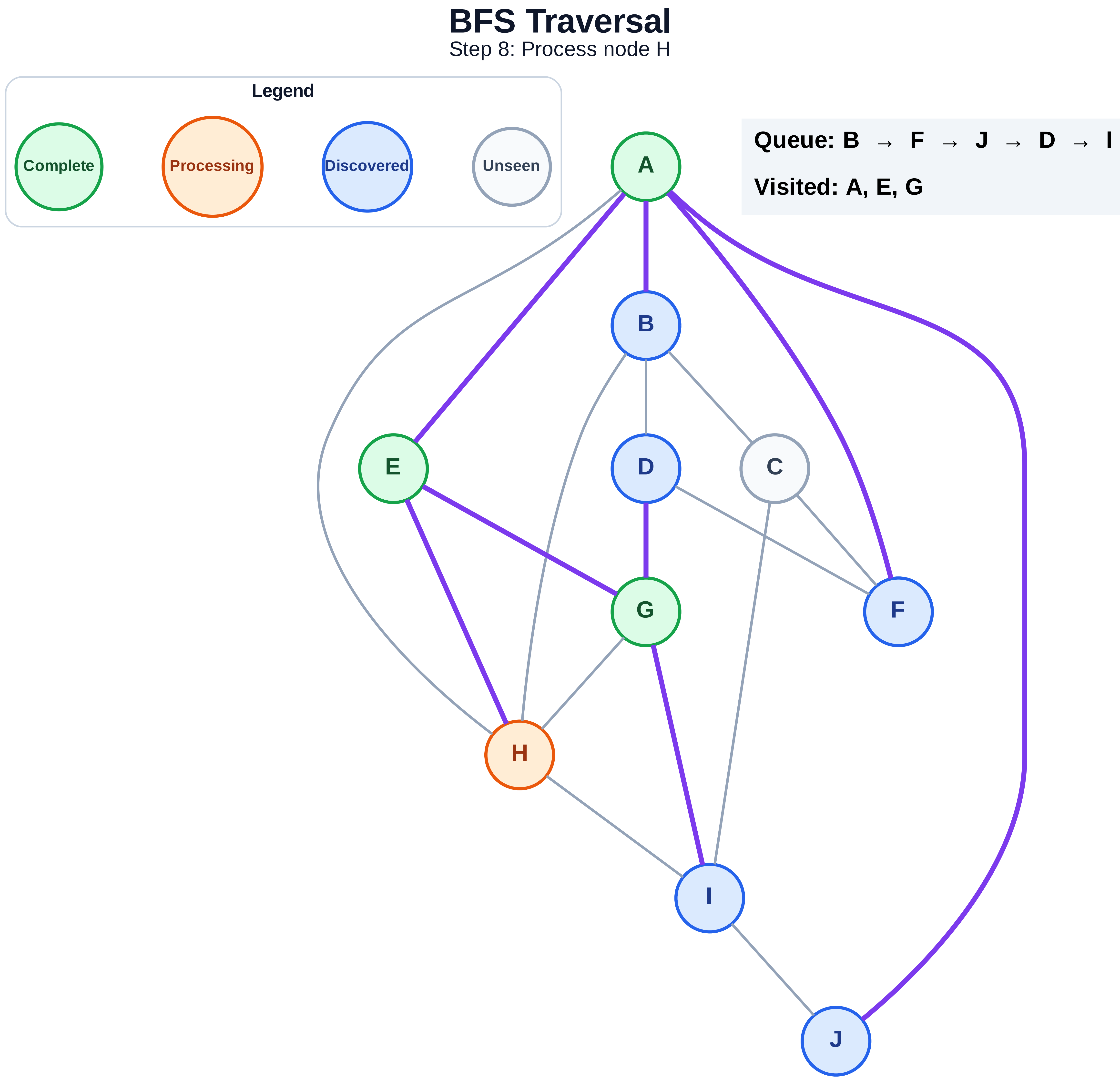
BFS Traversal

Step 6: Process node G



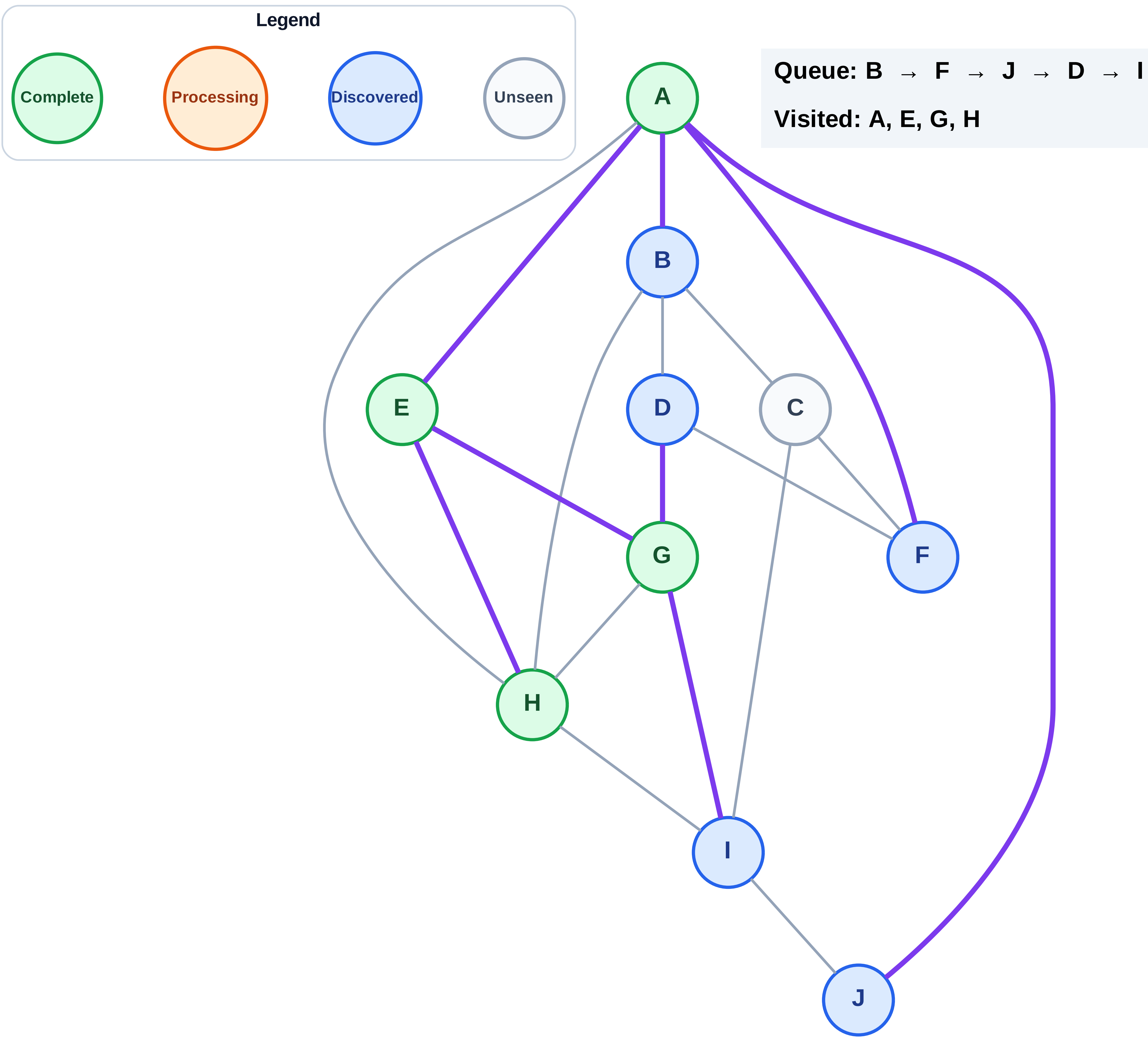


BFS Traversal
Step 7: Discover D, I from G
Legend
Complete
Processing
Discovered
Unseen
Queue: H → B → F → J → D → I
Visited: A, E, G
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 8: Process node H
Legend
Complete
Processing
Discovered
Unseen
Queue: B → F → J → D → I
Visited: A, E, G
A
B
E
D
C
G
F
H
I
J

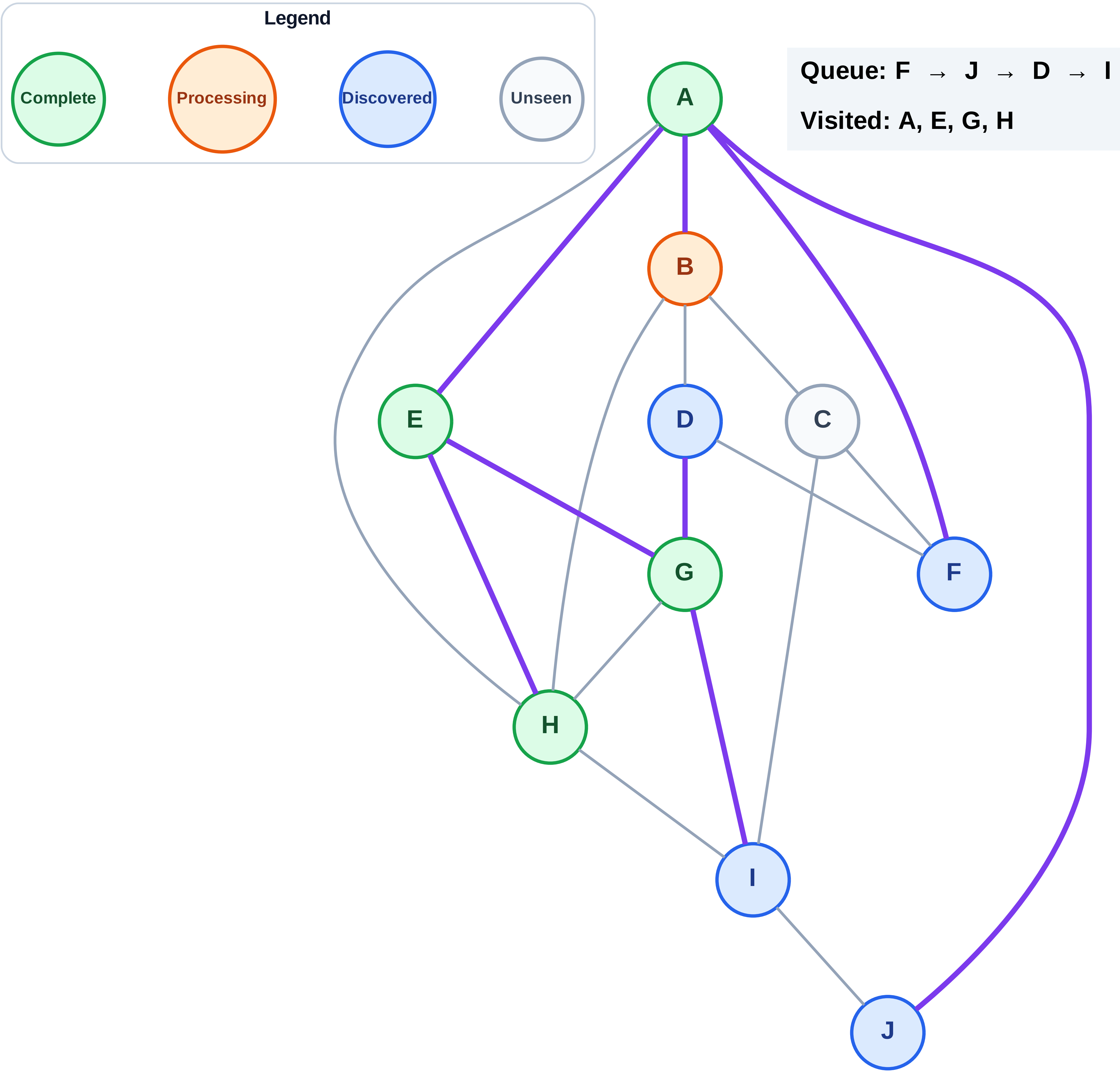
BFS Traversal

Step 9: No new neighbors from H



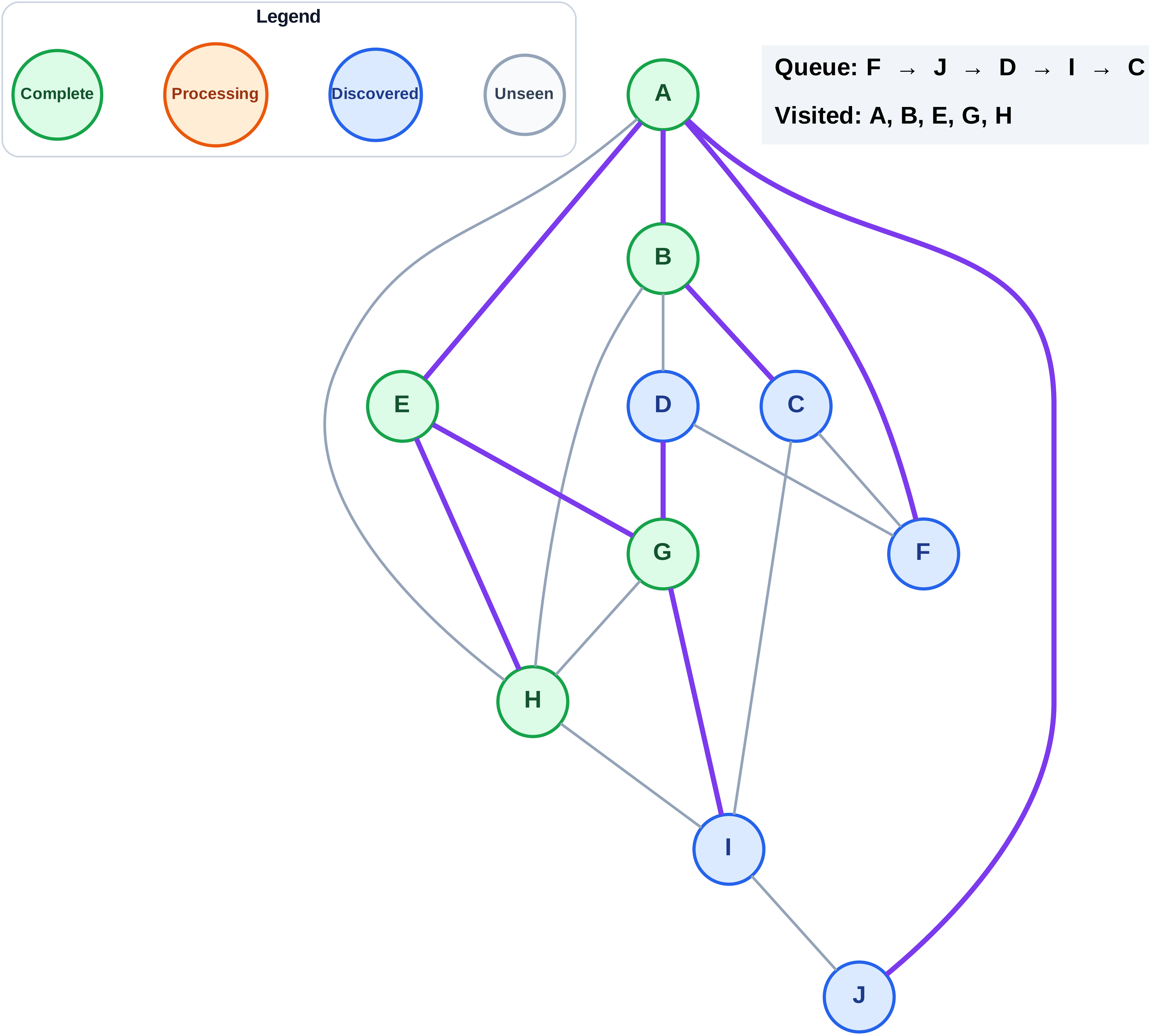
BFS Traversal

Step 10: Process node B



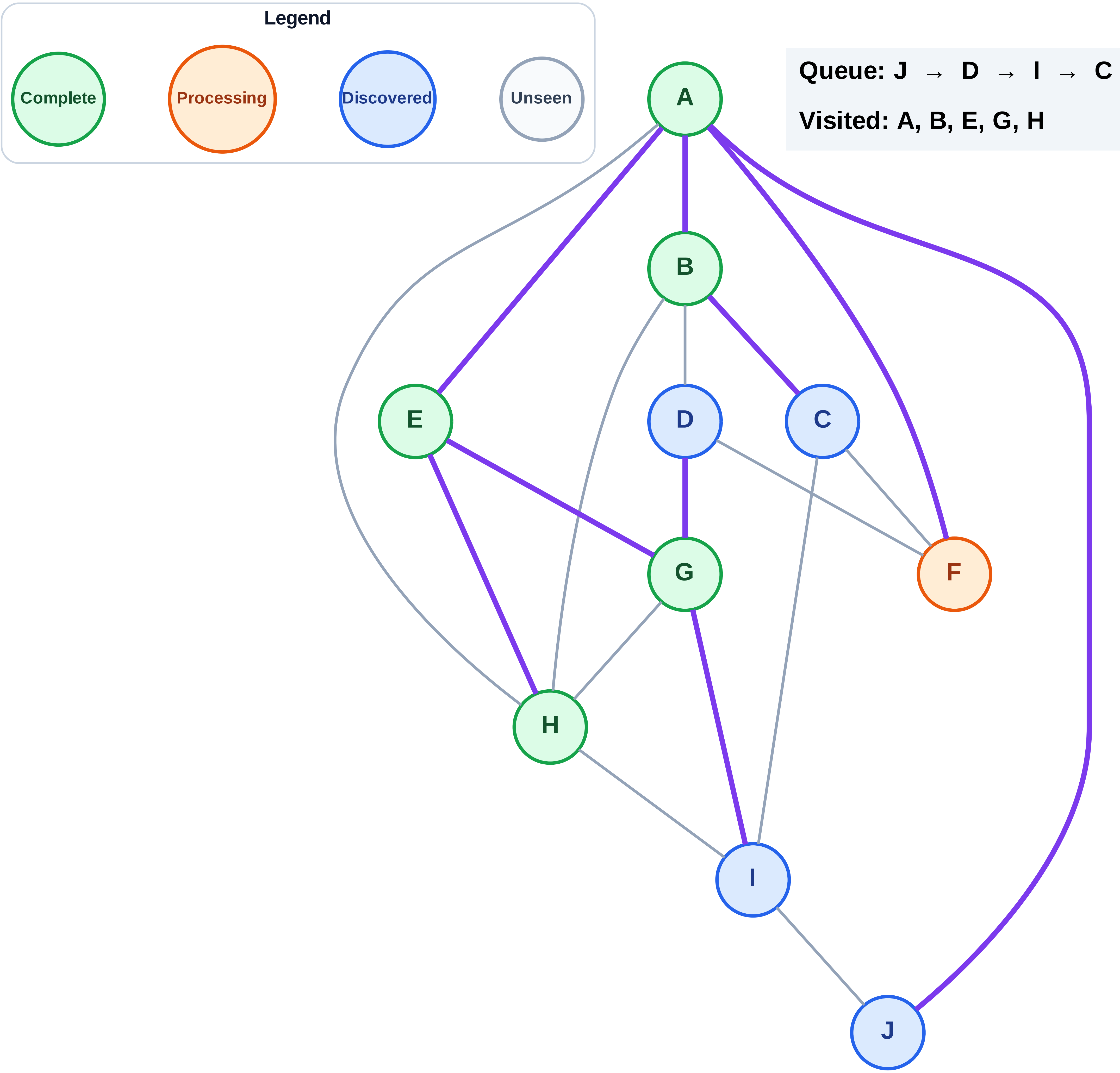
BFS Traversal

Step 11: Discover C from B



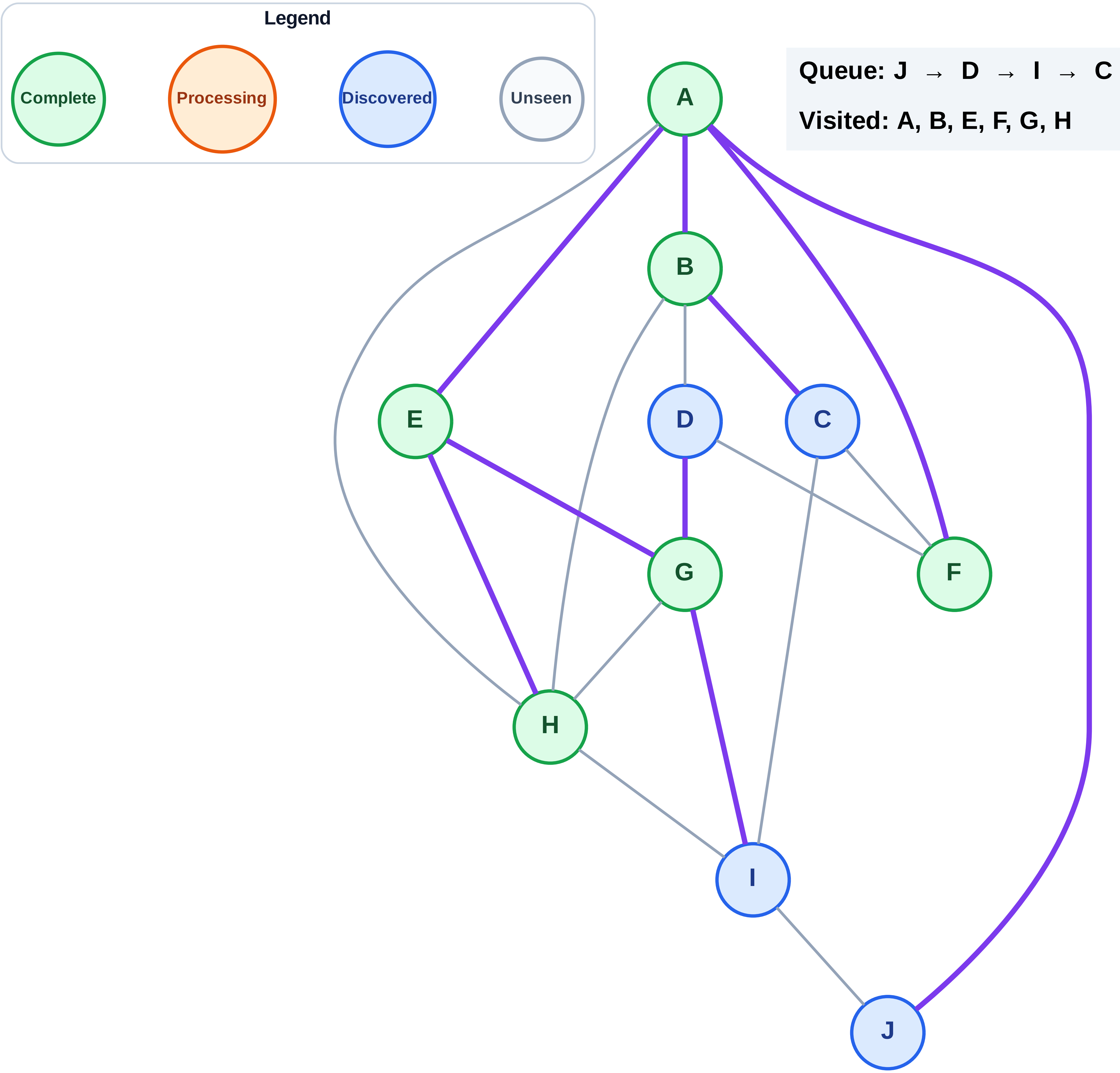
BFS Traversal

Step 12: Process node F



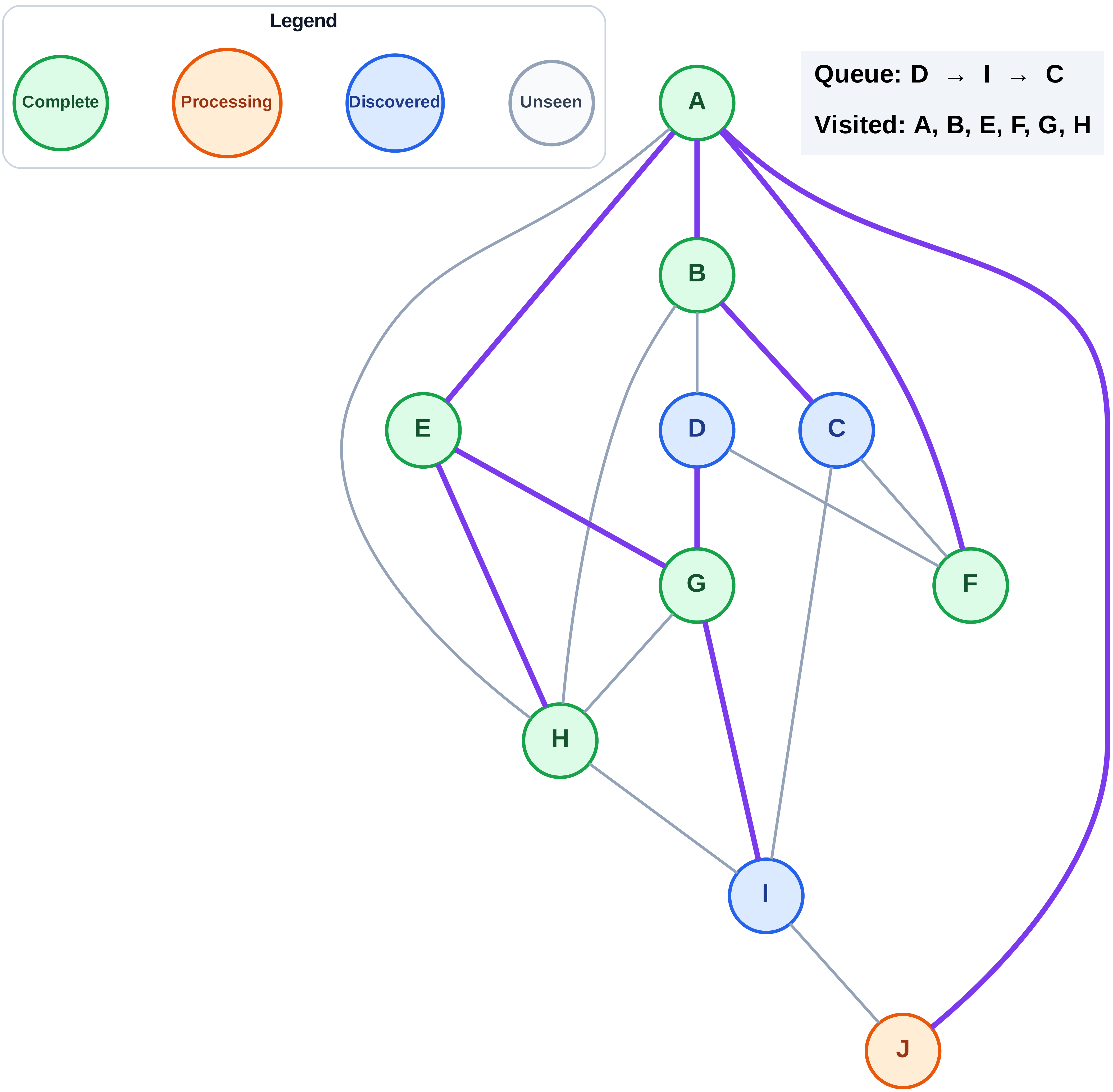
BFS Traversal

Step 13: No new neighbors from F



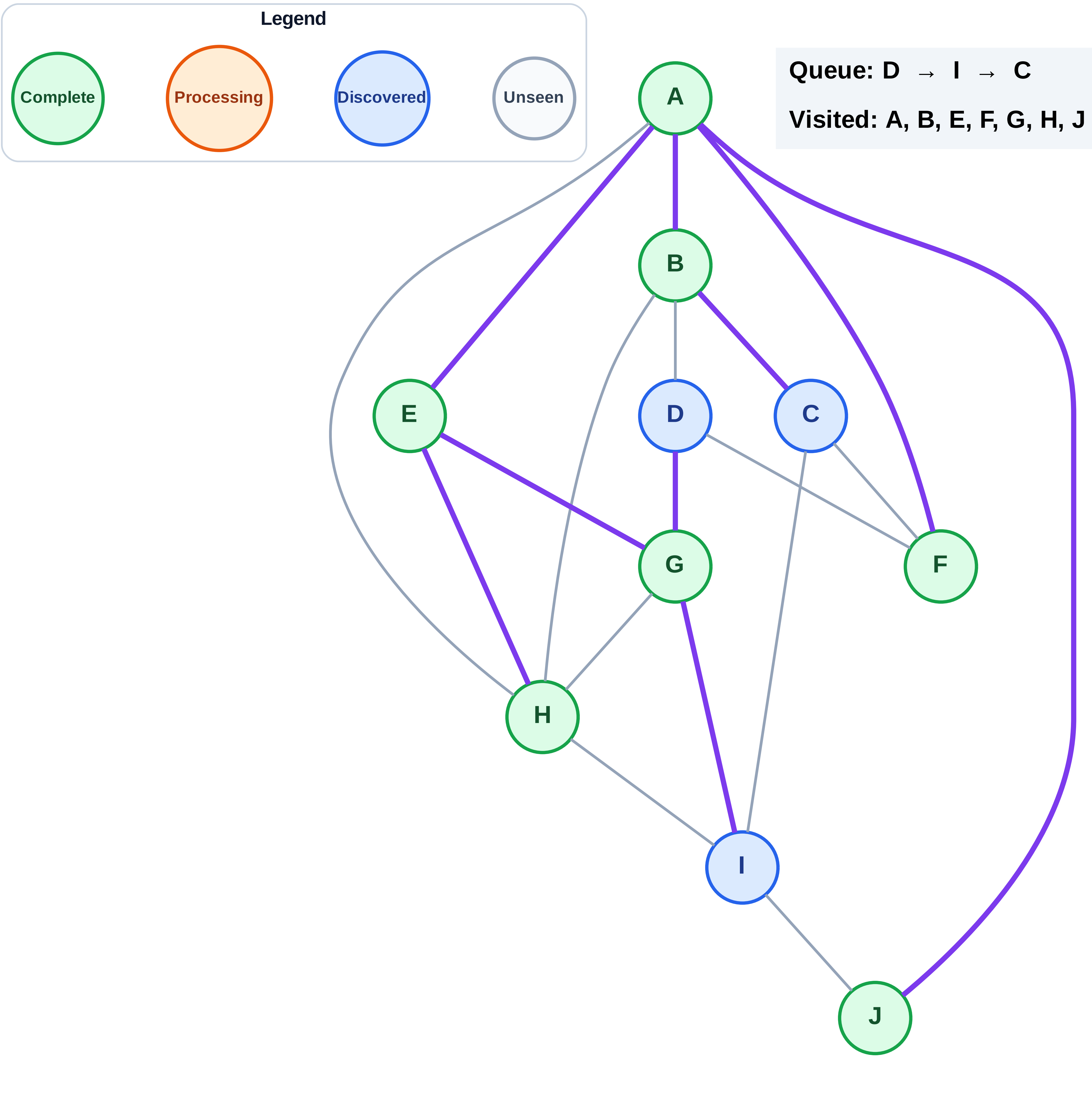
BFS Traversal

Step 14: Process node J



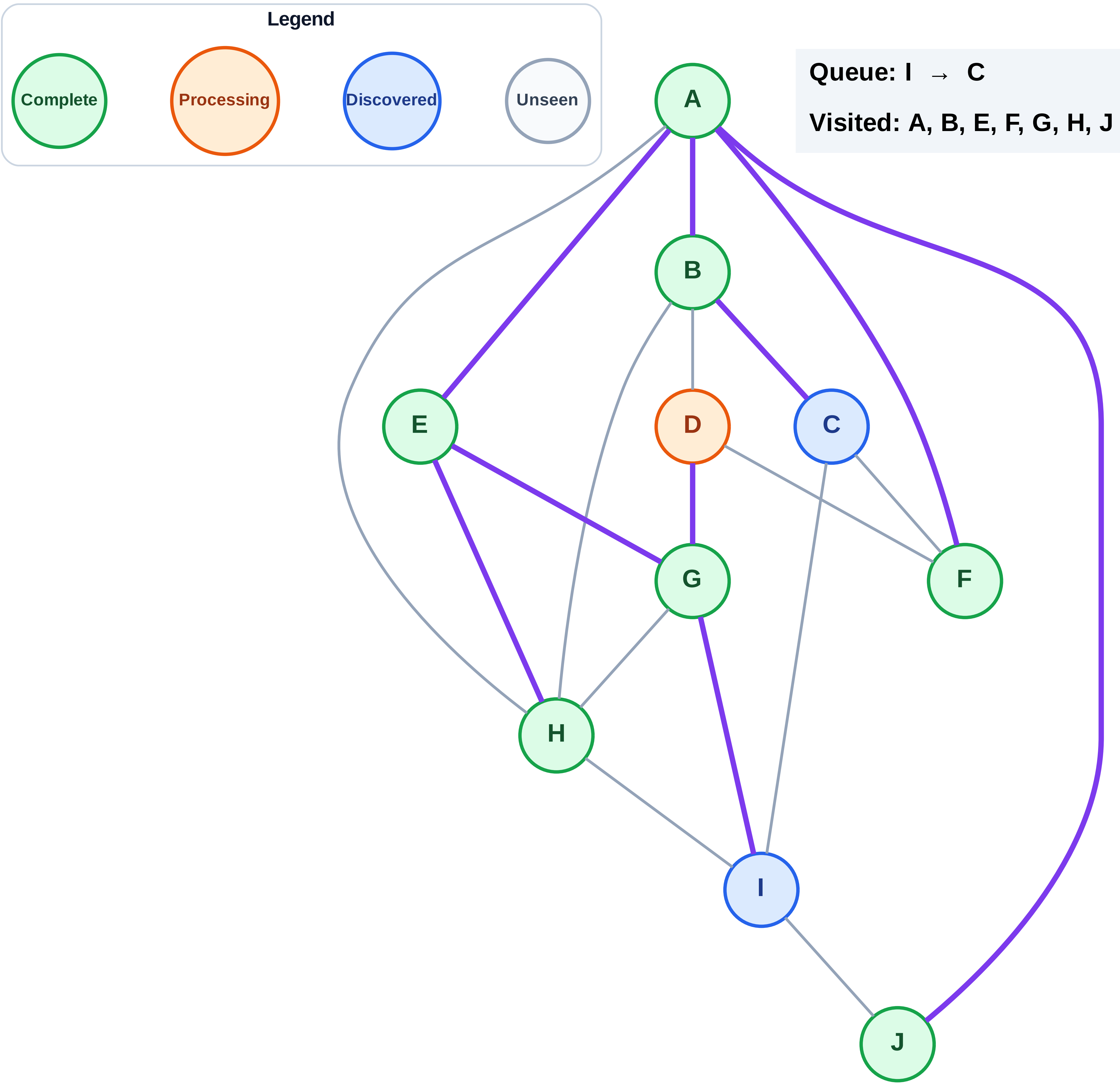
BFS Traversal

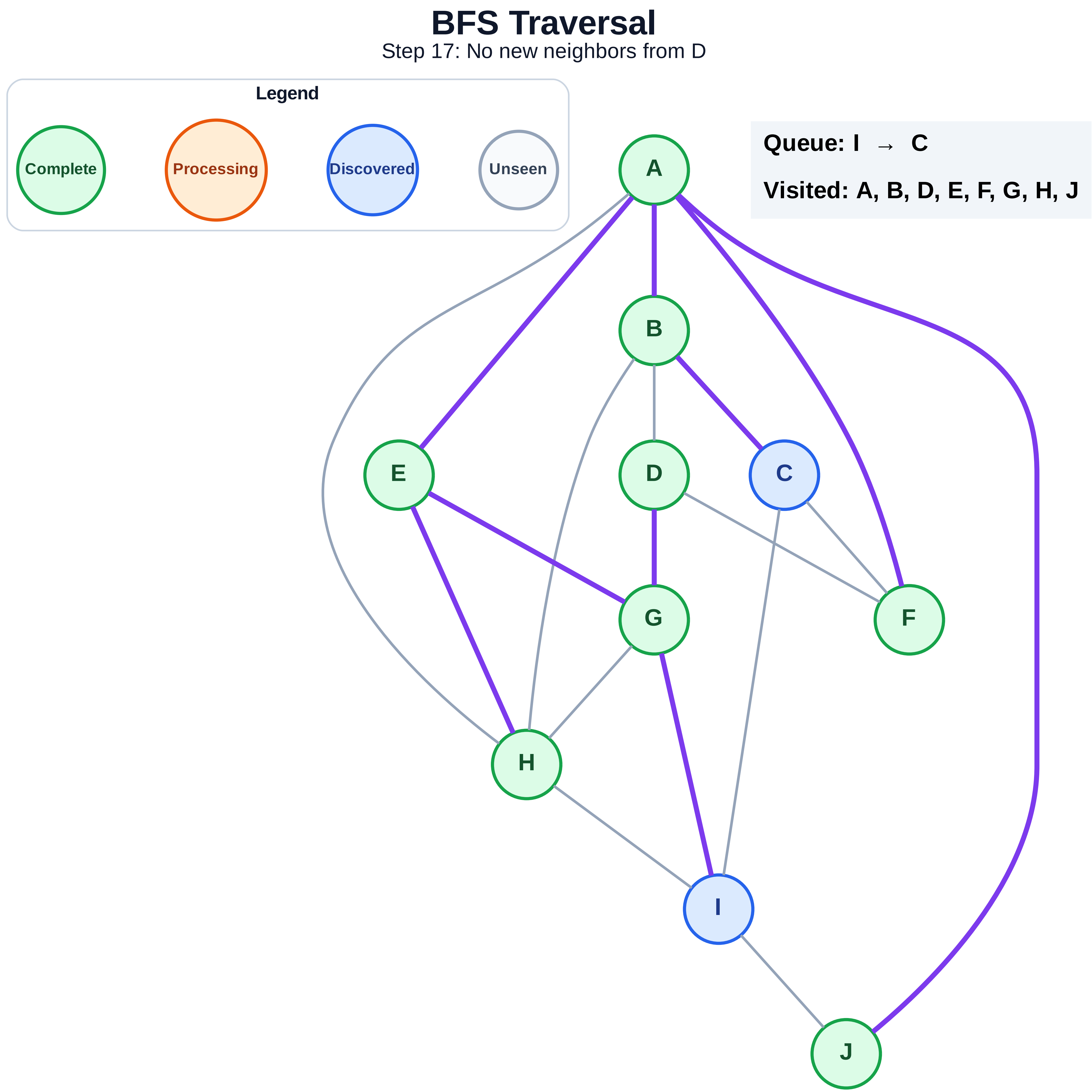
Step 15: No new neighbors from J



BFS Traversal

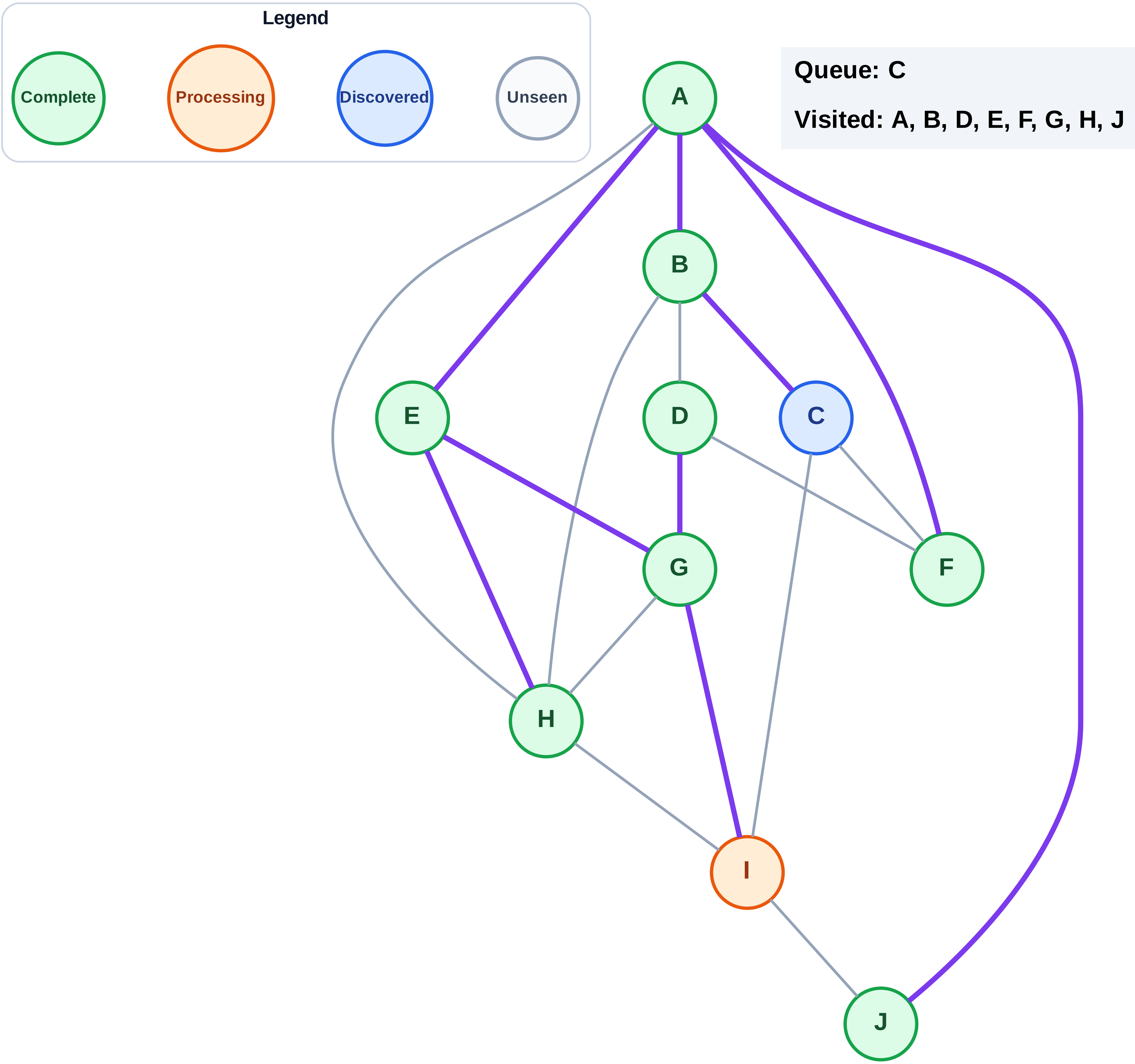
Step 16: Process node D




BFS Traversal
Step 17: No new neighbors from D
Legend
Complete
Processing
Discovered
Unseen
Queue: I → C
Visited: A, B, D, E, F, G, H, J
A
B
E
D
C
G
F
H
I
J

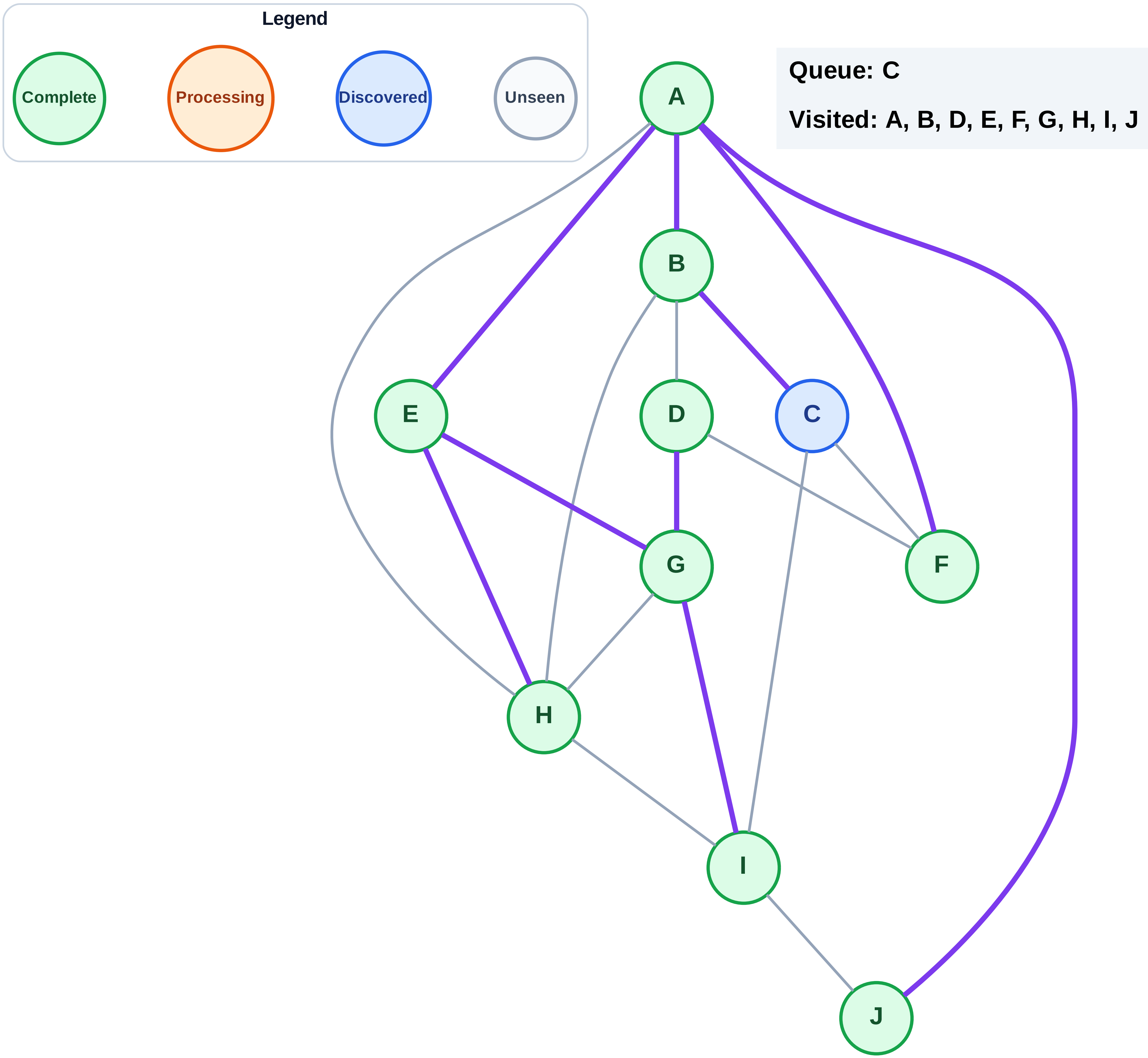
BFS Traversal

Step 18: Process node I



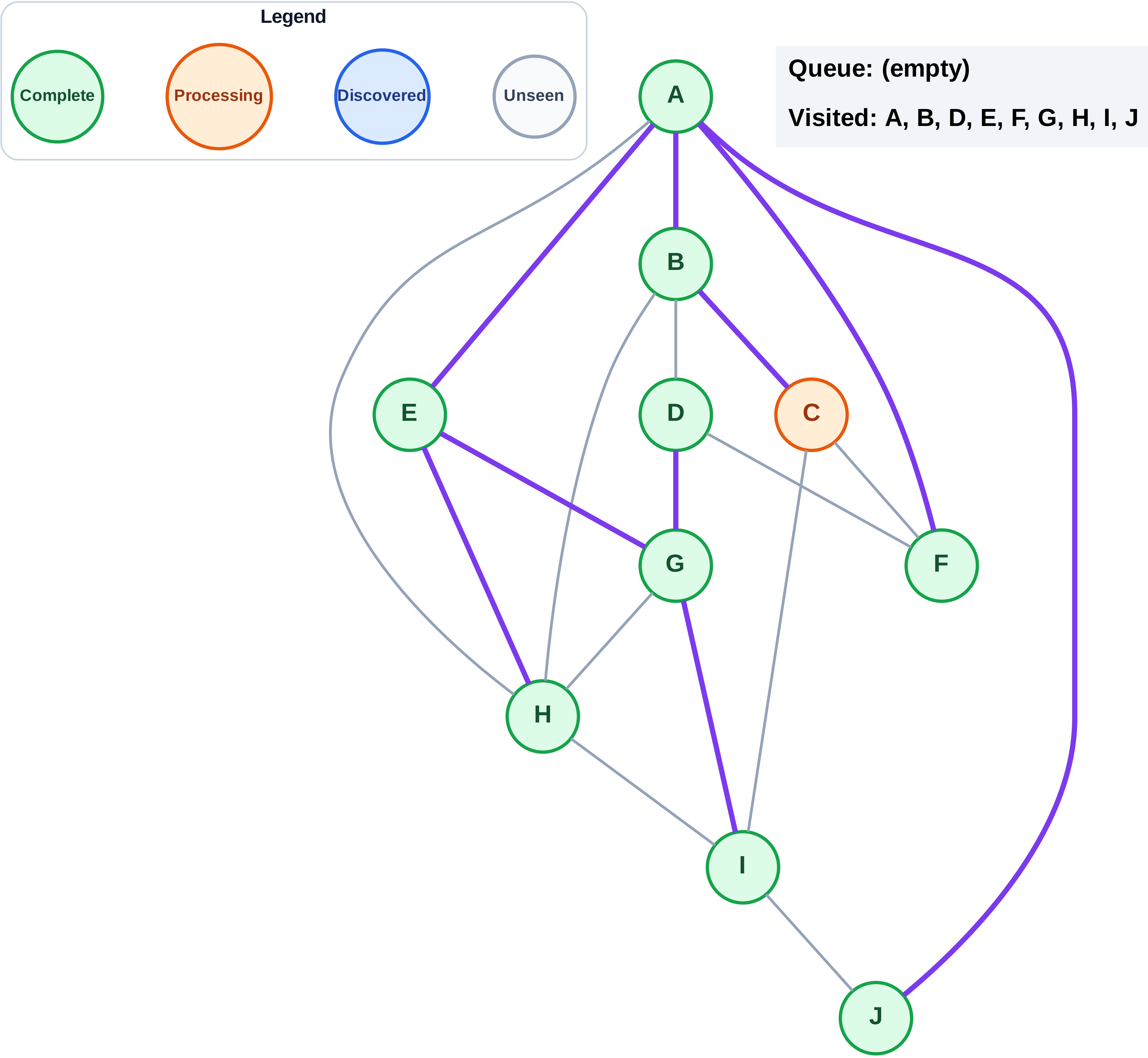
BFS Traversal

Step 19: No new neighbors from I



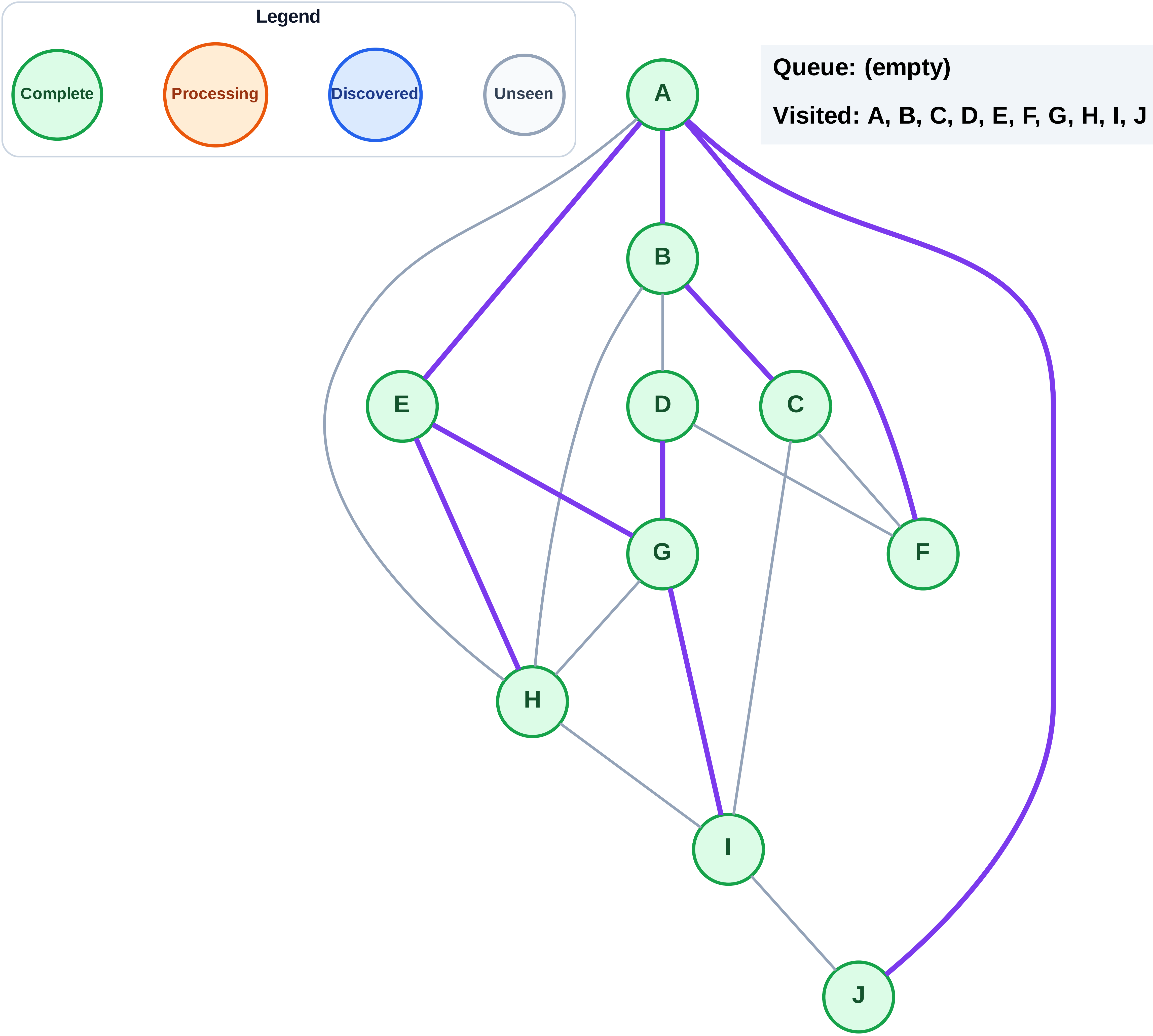
BFS Traversal

Step 20: Process node C



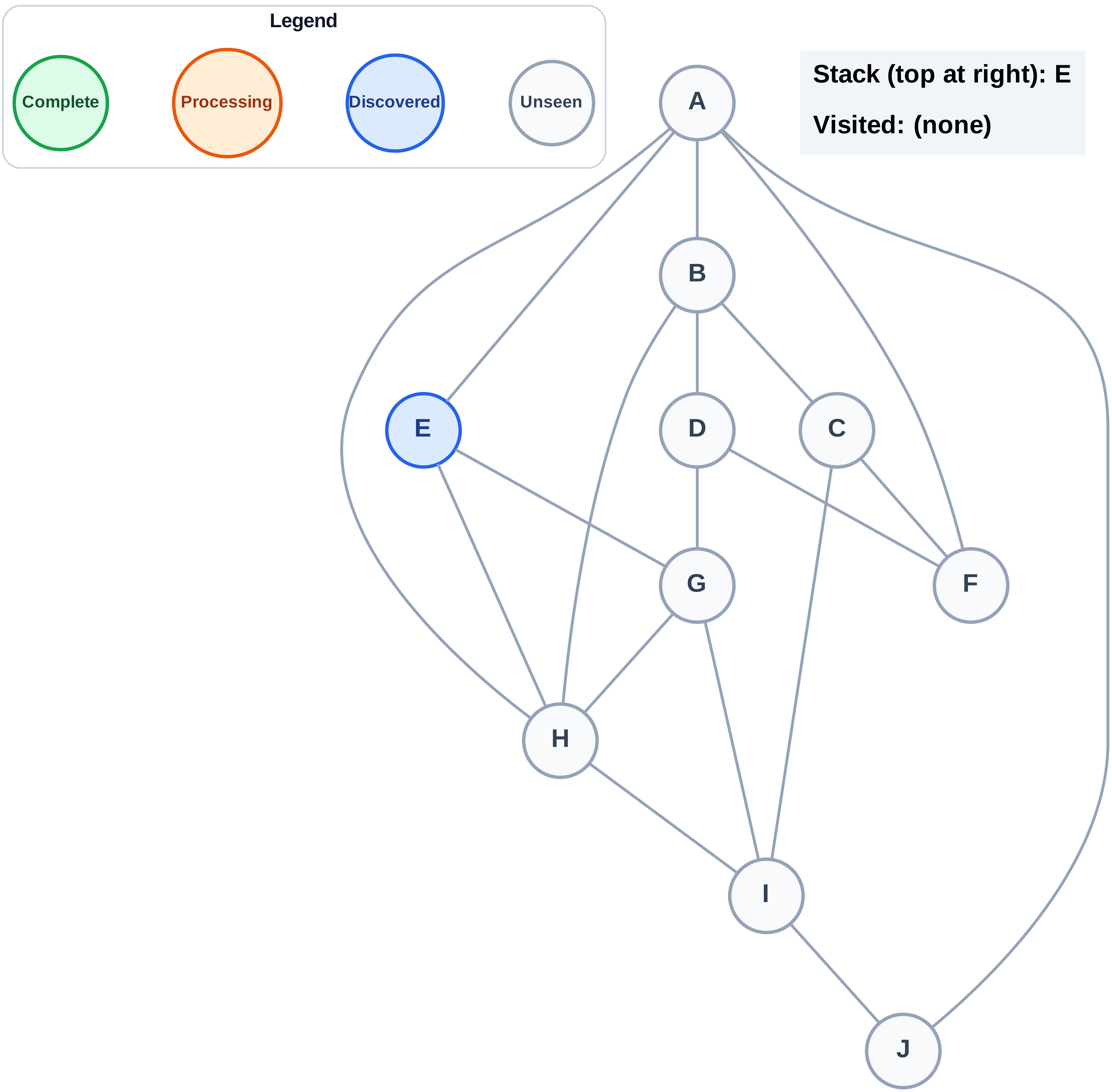
BFS Traversal

Step 21: No new neighbors from C



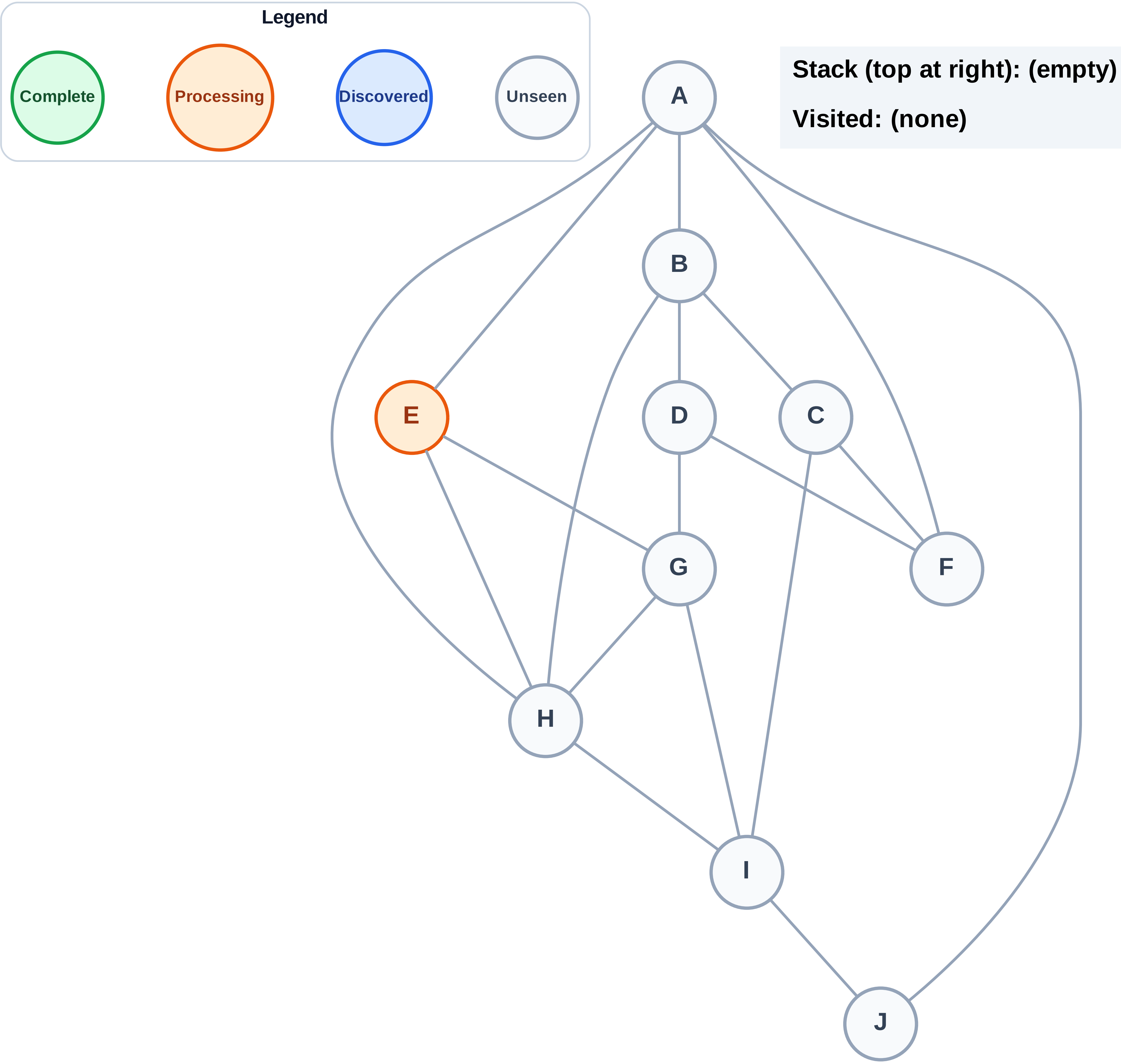
DFS Traversal

Step 1: Start at E



DFS Traversal

Step 2: Process node E

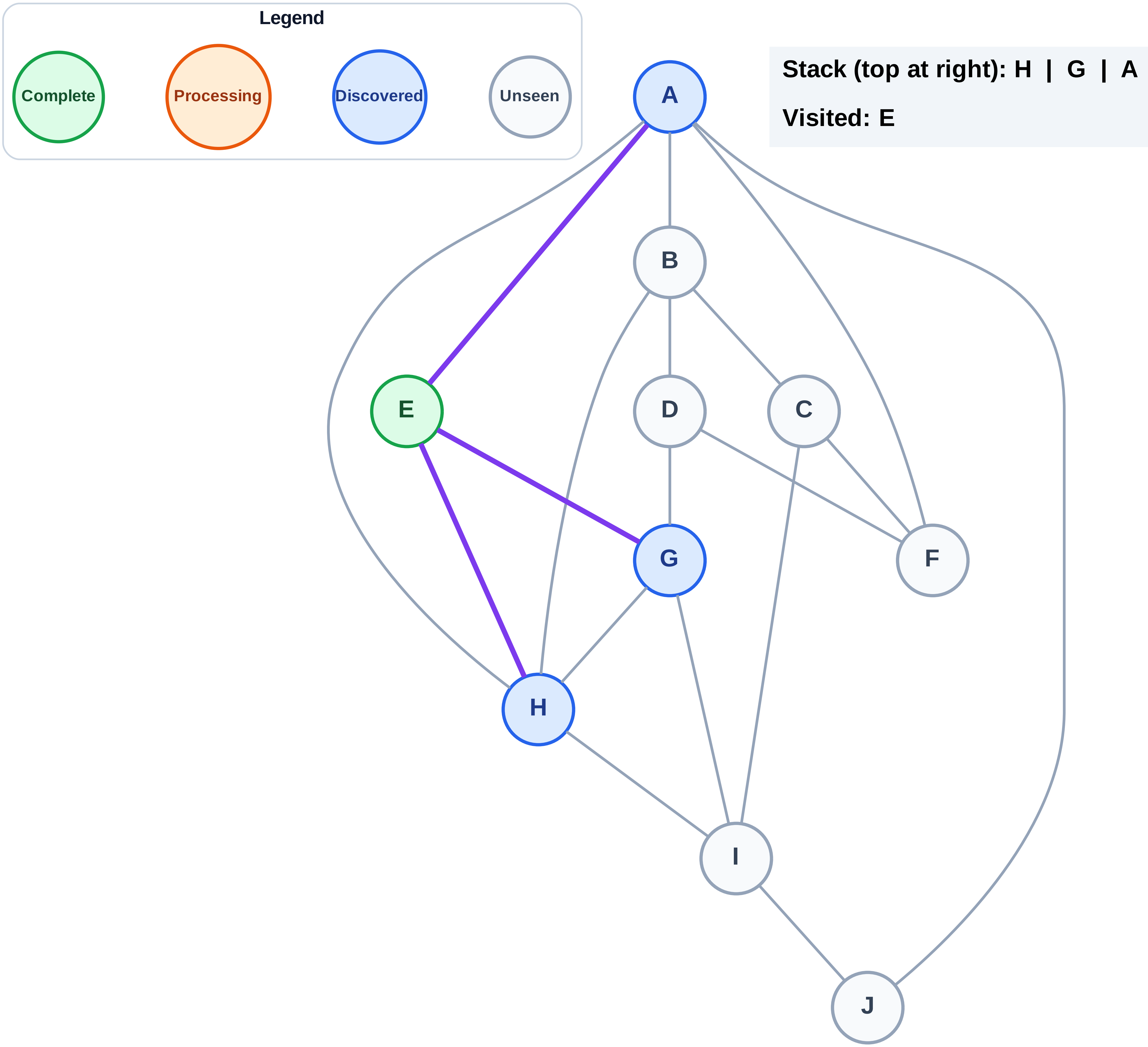


Stack (top at right): (empty)

Visited: (none)

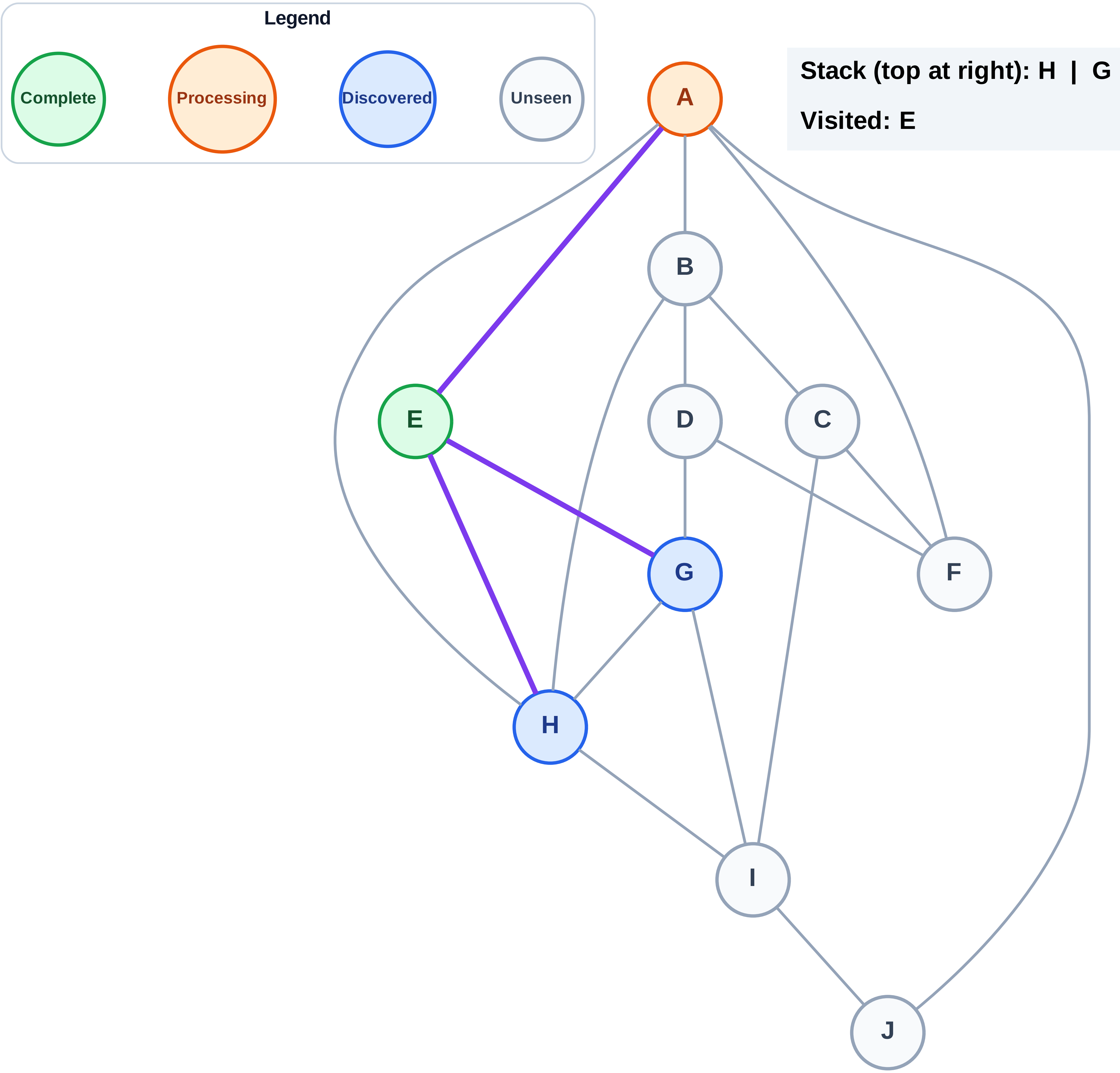
DFS Traversal

Step 3: Discover H, G, A from E



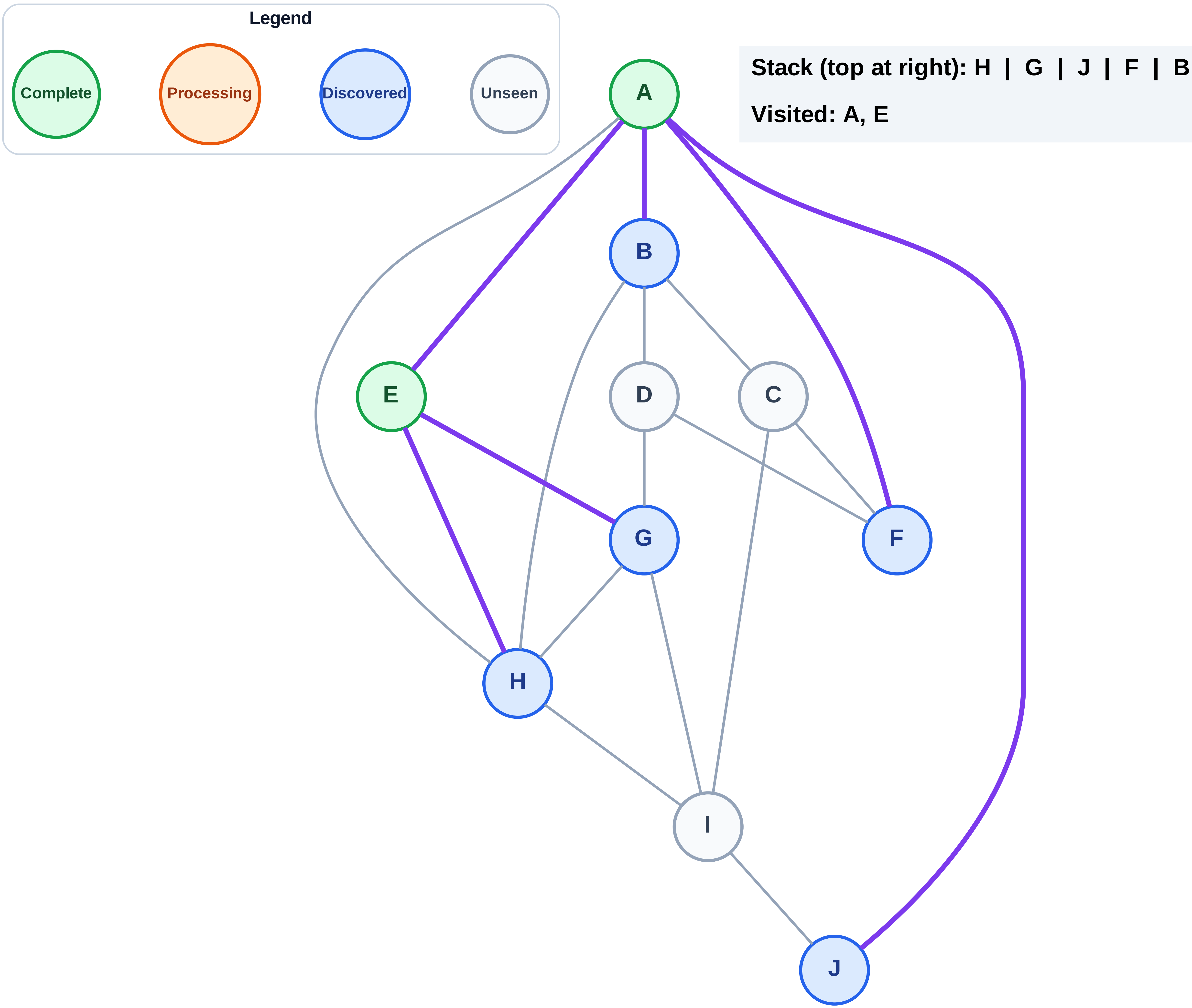
DFS Traversal

Step 4: Process node A



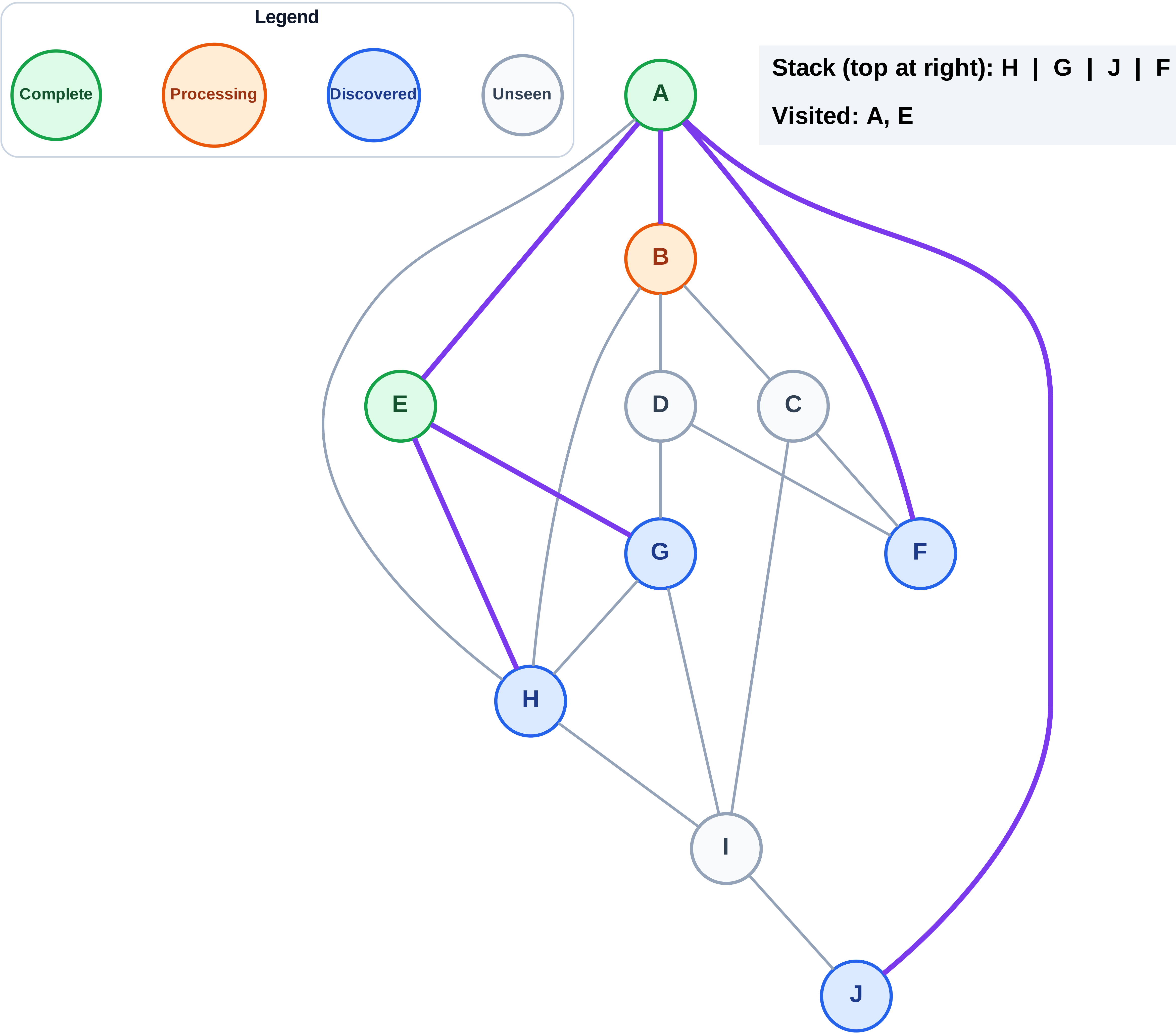
DFS Traversal

Step 5: Discover J, F, B from A



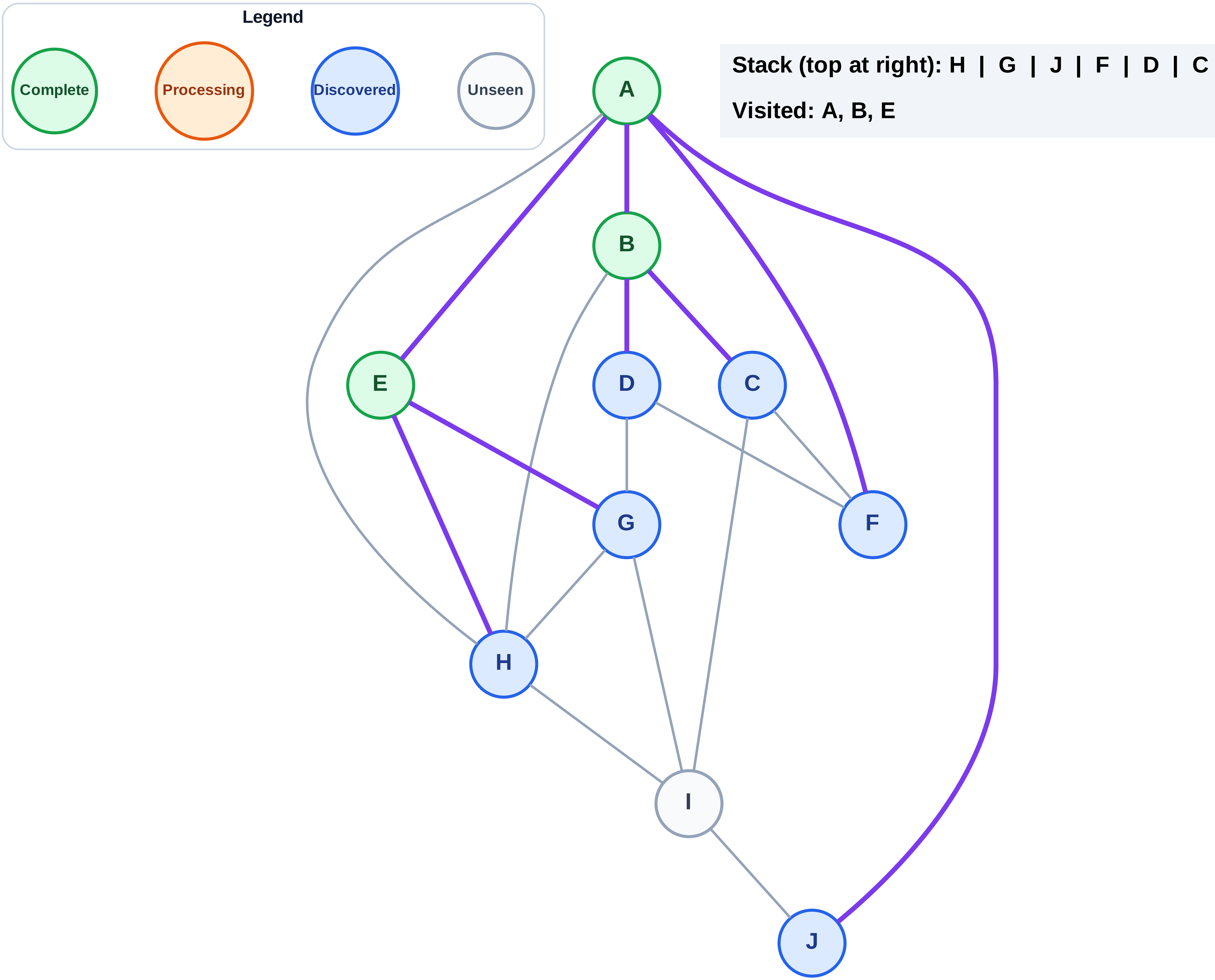
DFS Traversal

Step 6: Process node B



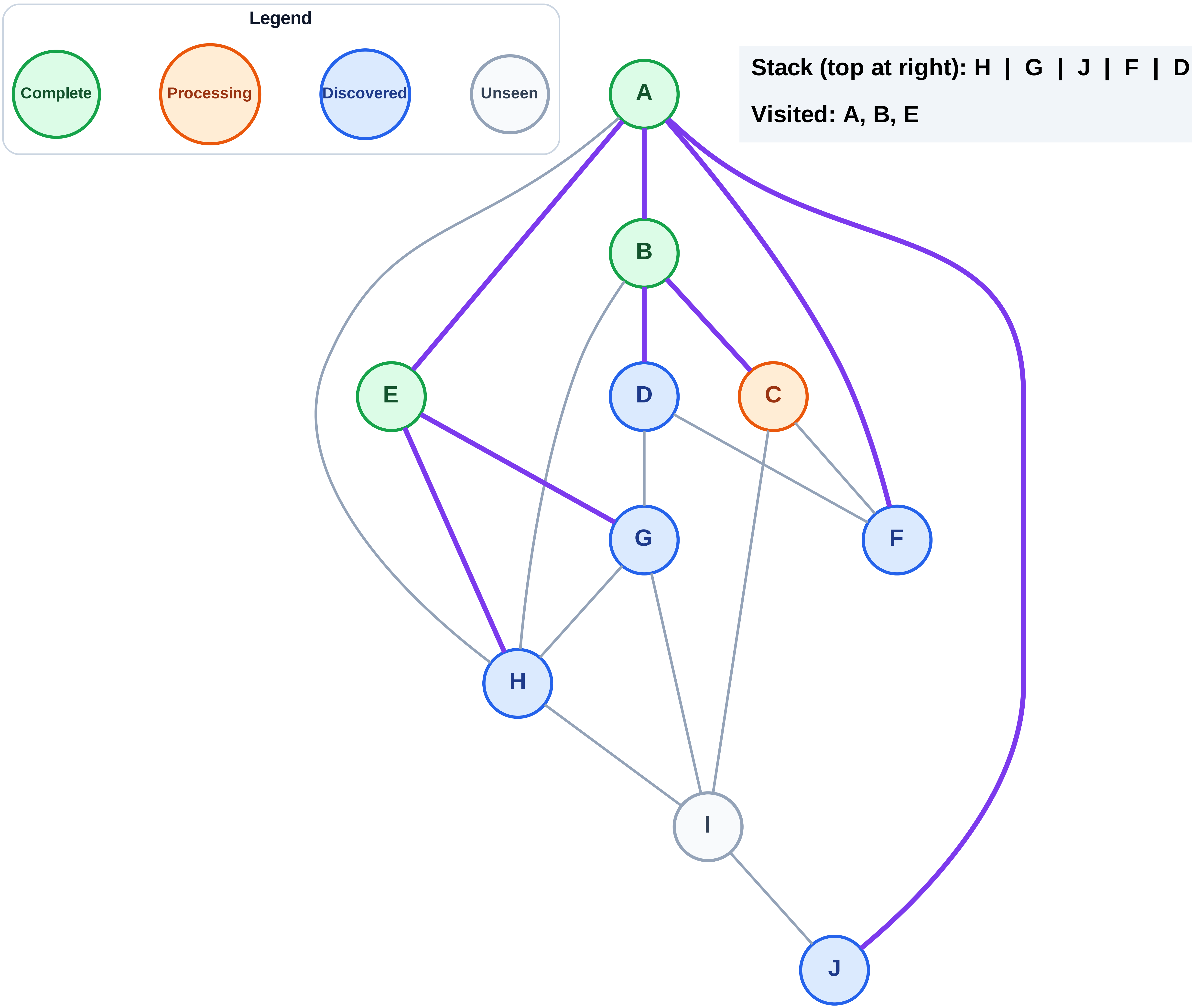
DFS Traversal

Step 7: Discover D, C from B



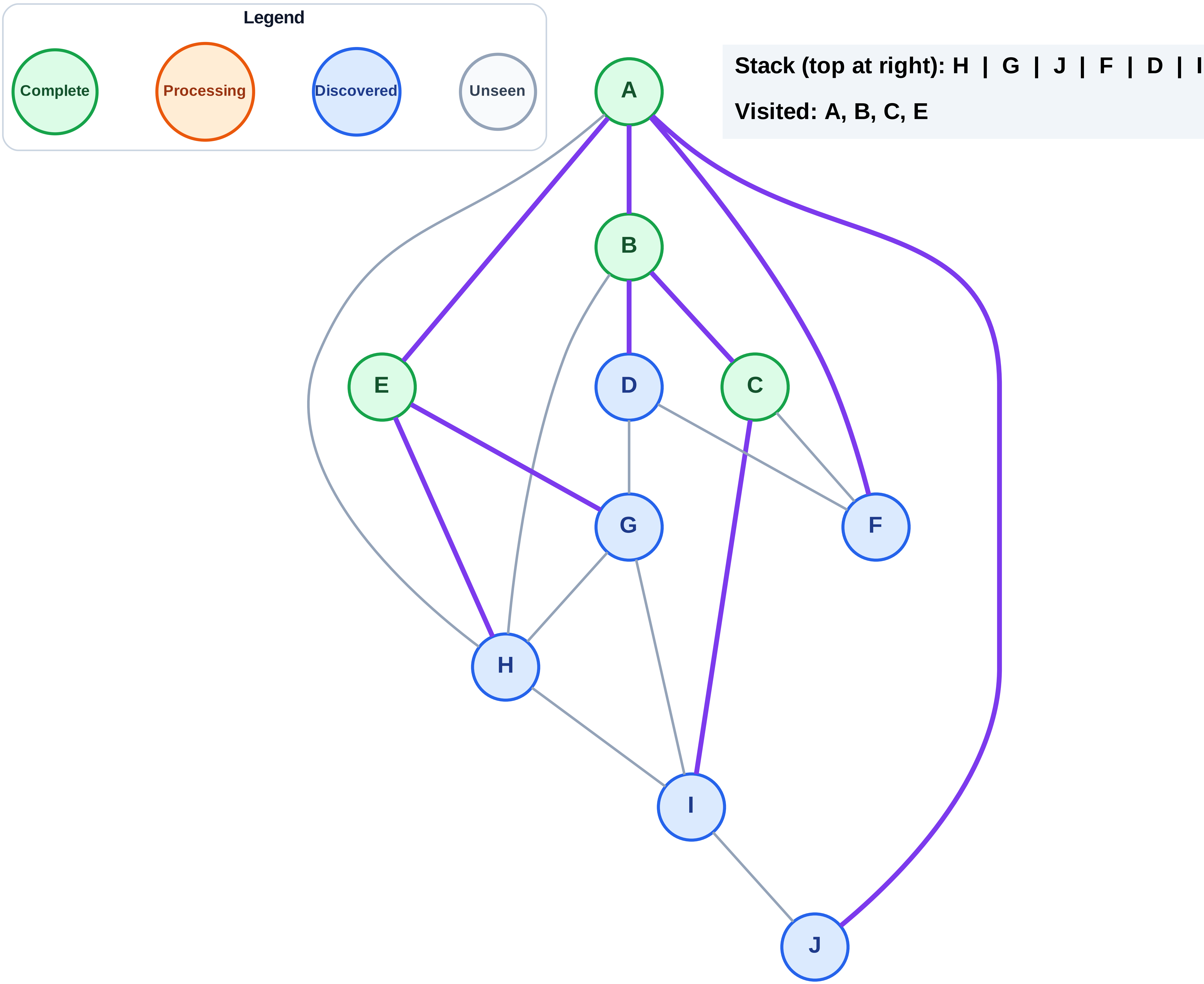
DFS Traversal

Step 8: Process node C



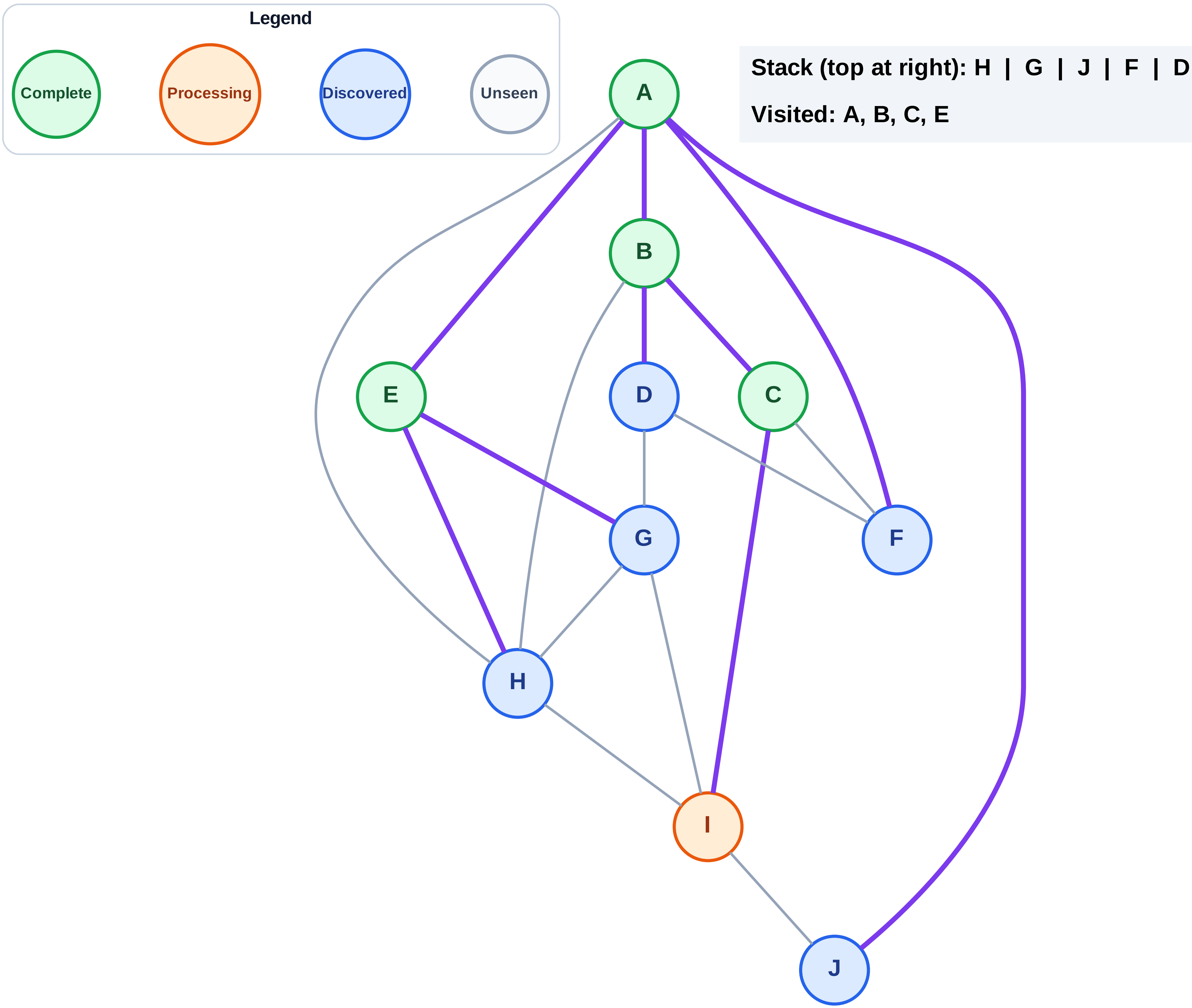
DFS Traversal

Step 9: Discover I from C



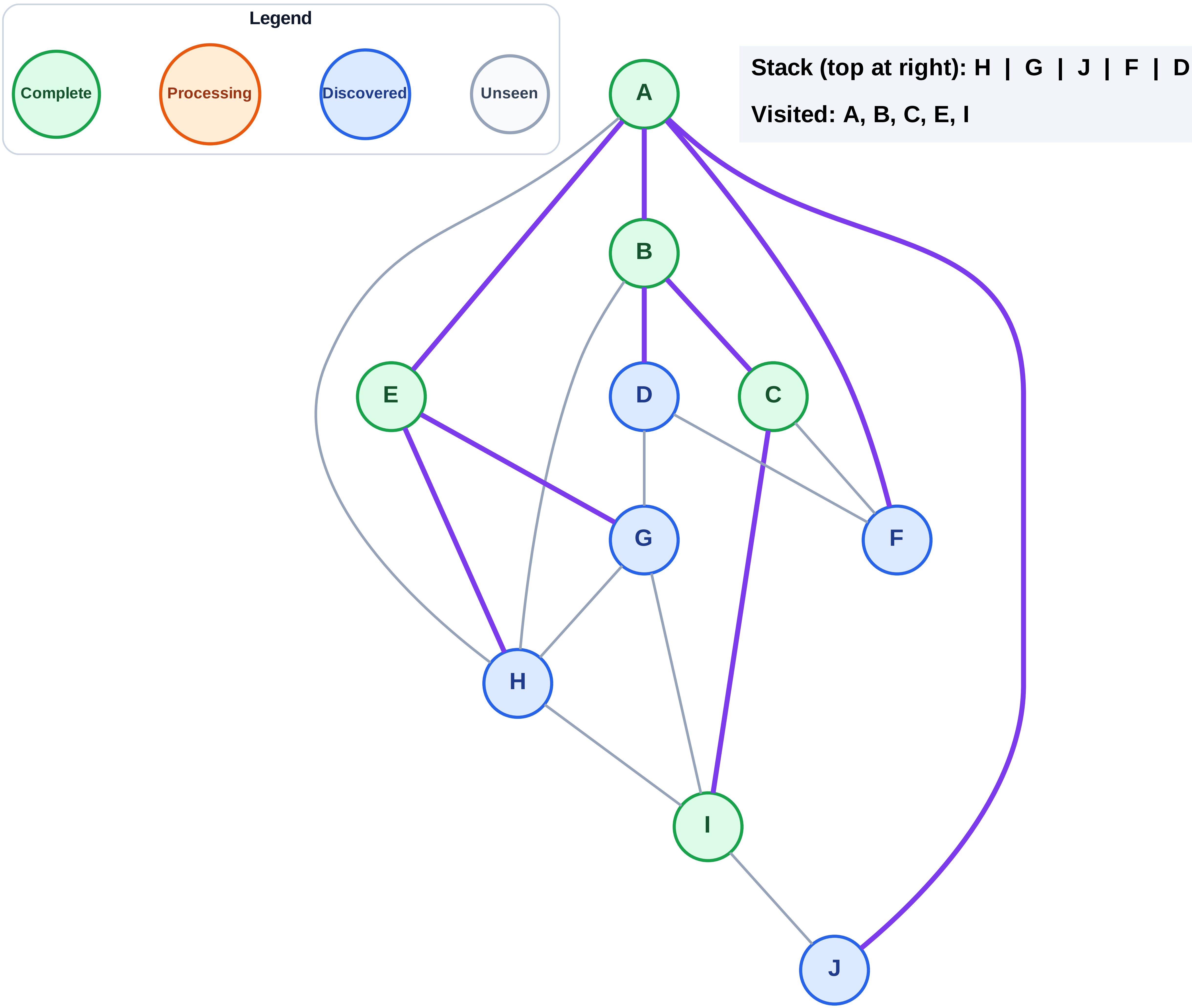
DFS Traversal

Step 10: Process node I



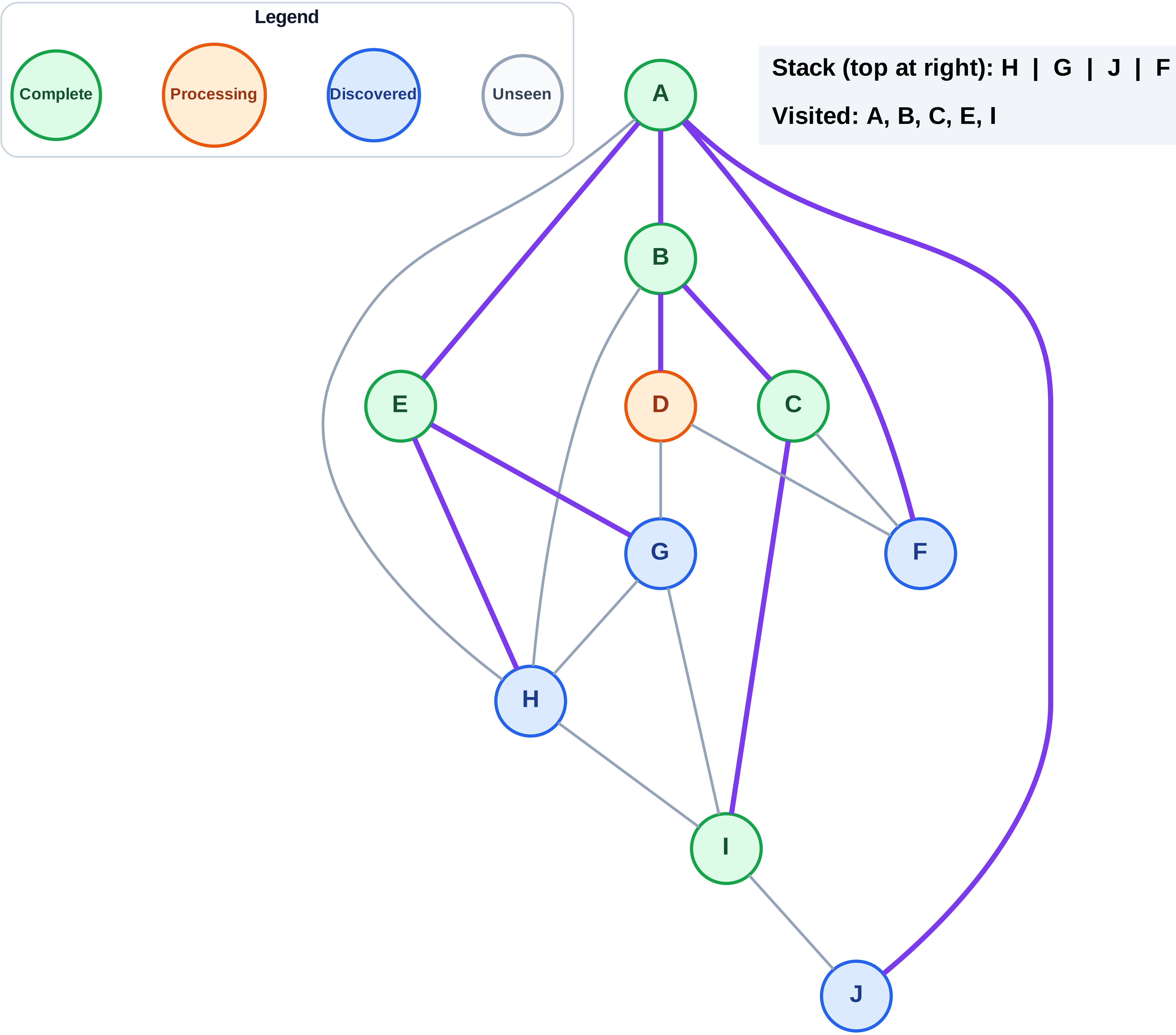
DFS Traversal

Step 11: No new neighbors from I



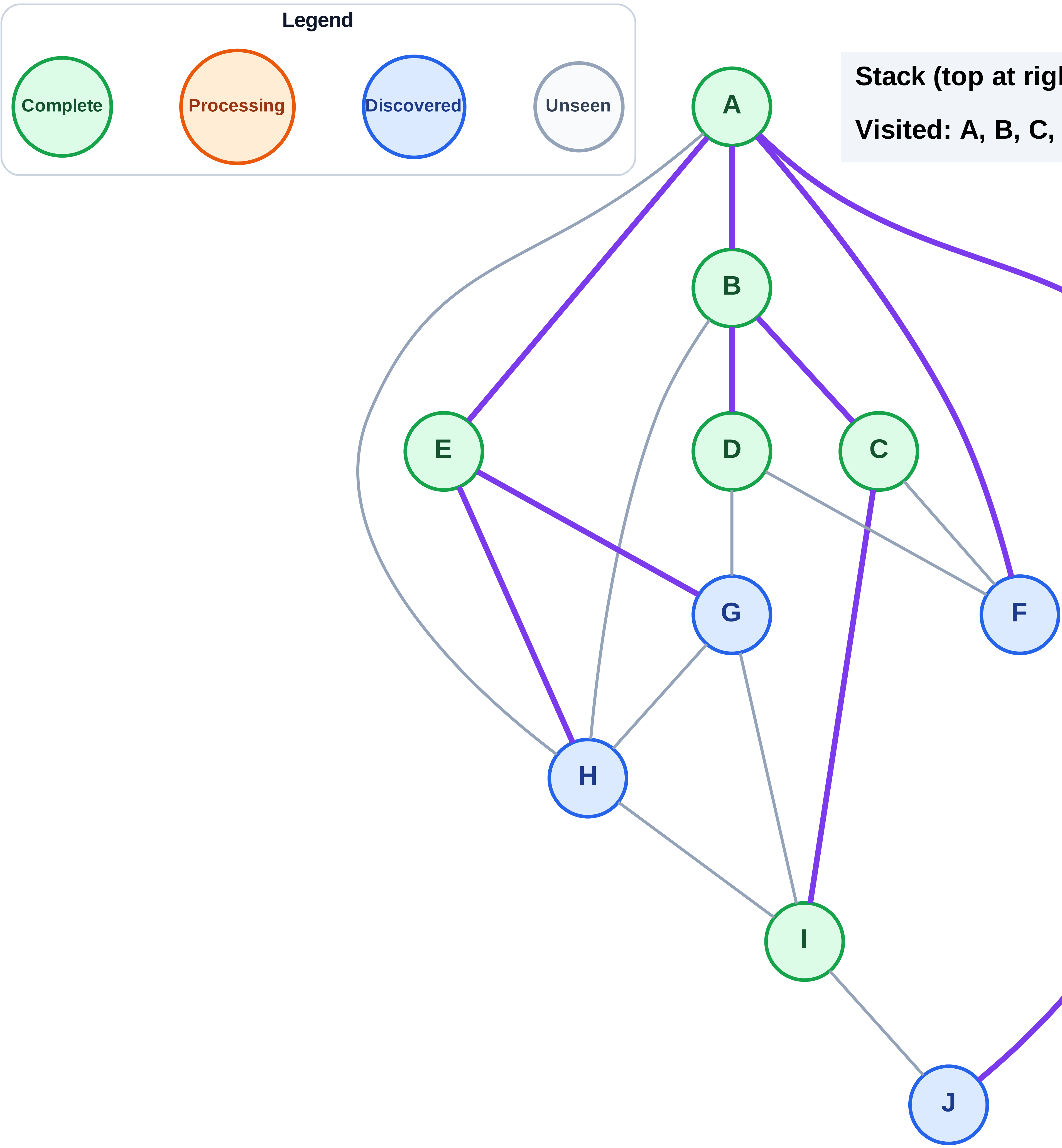
DFS Traversal

Step 12: Process node D



DFS Traversal

Step 13: No new neighbors from D

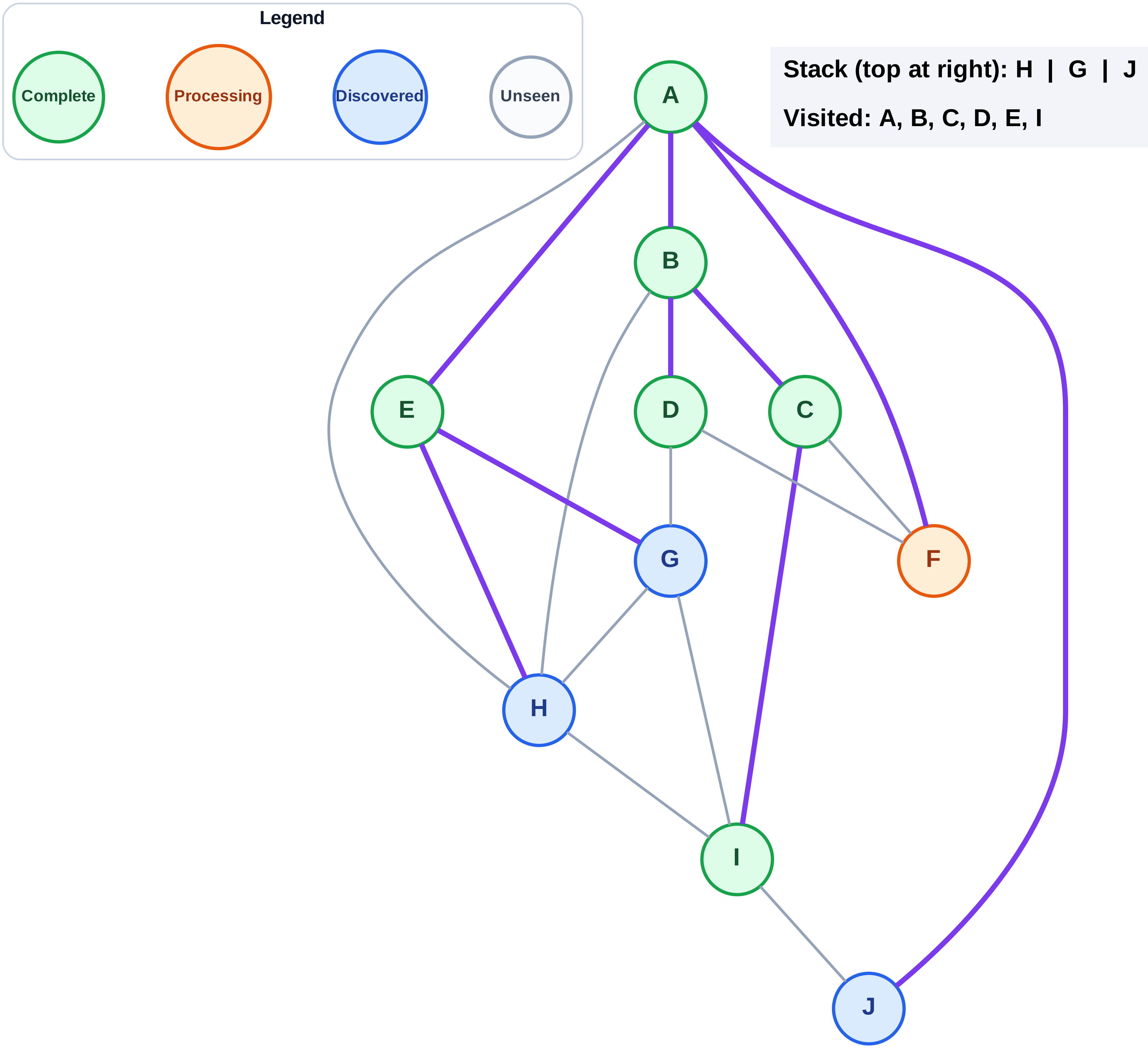


Stack (top at right): H | G | J | F

Visited: A, B, C, D, E, I

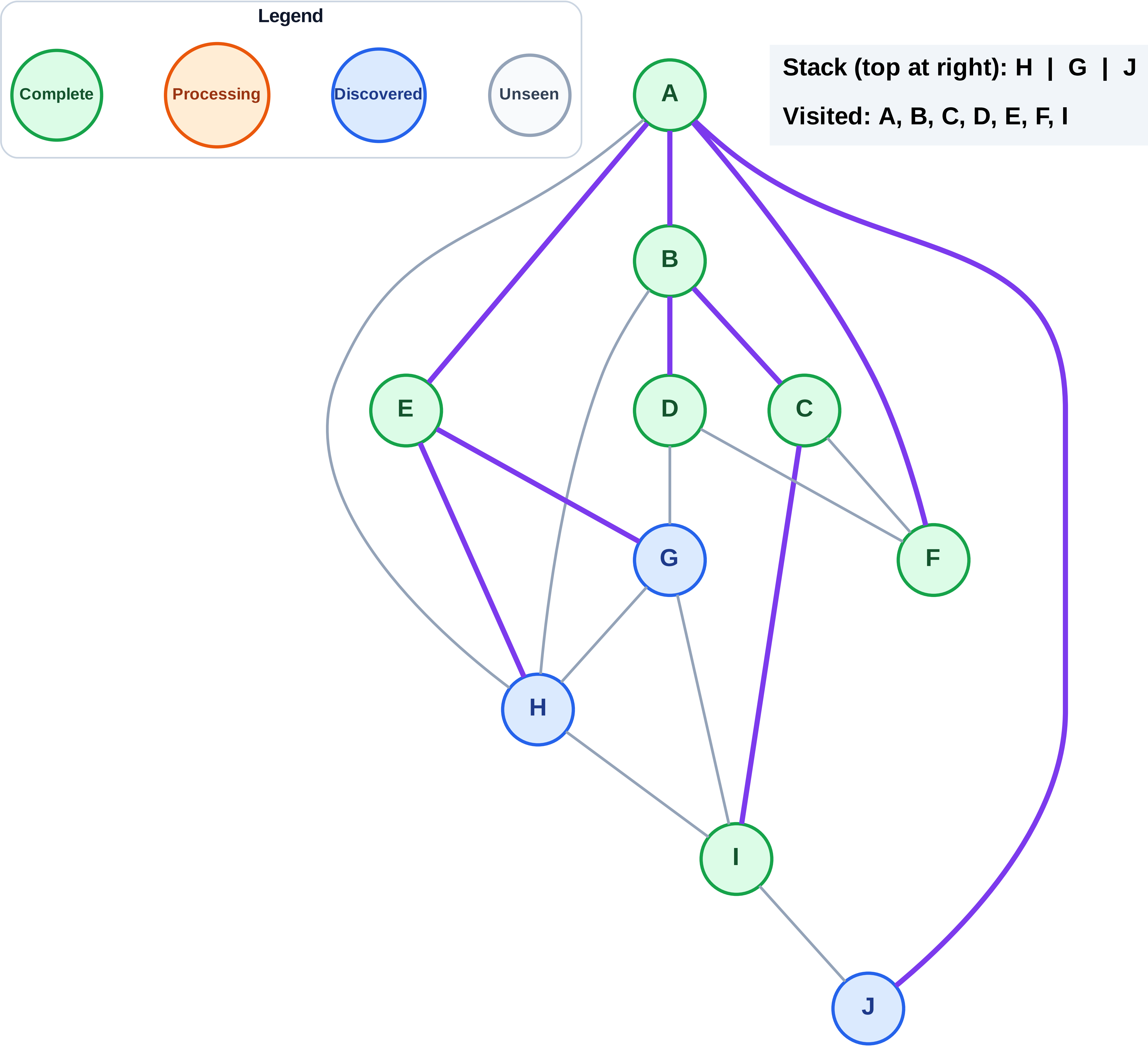
DFS Traversal

Step 14: Process node F



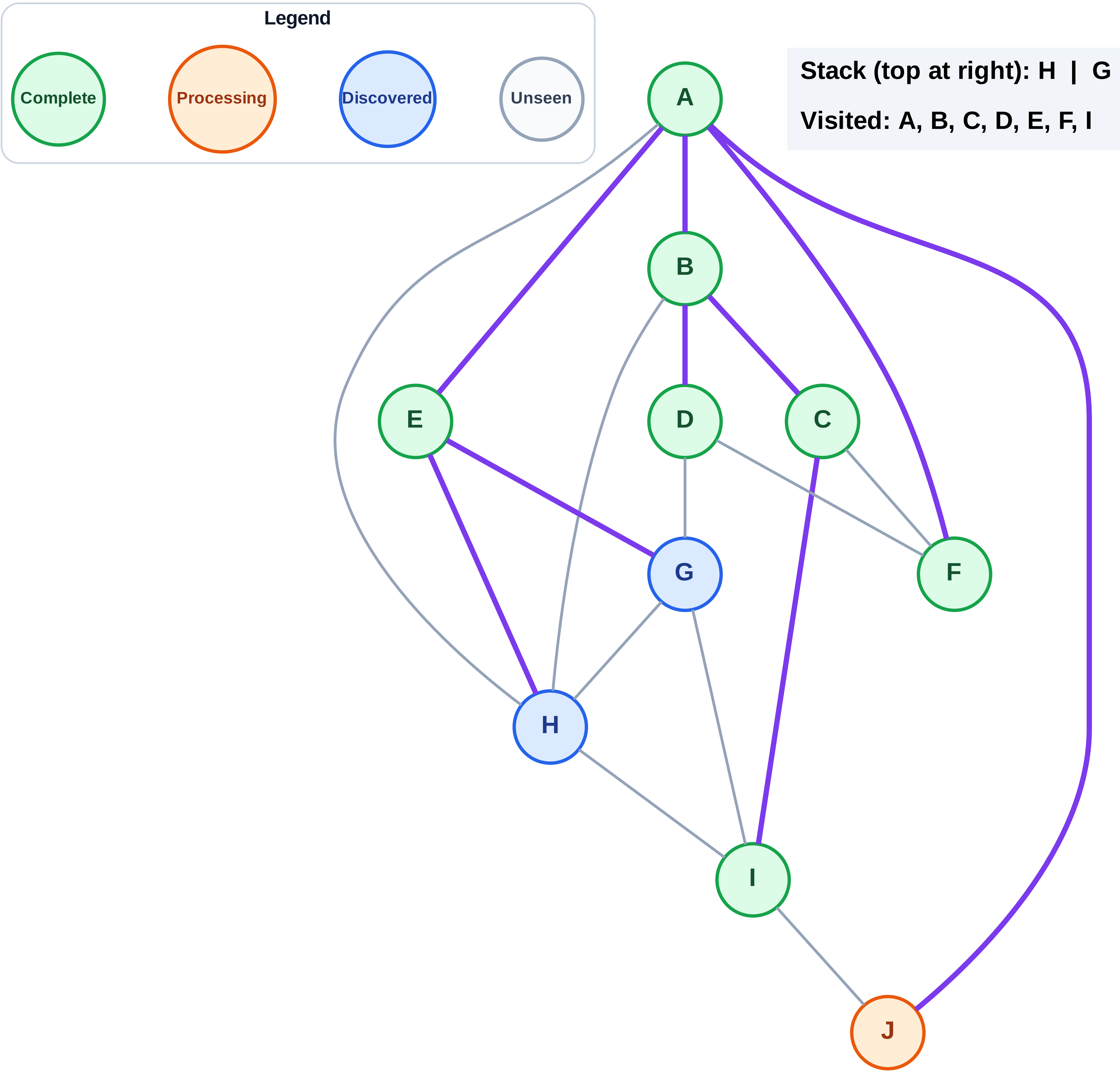
DFS Traversal

Step 15: No new neighbors from F



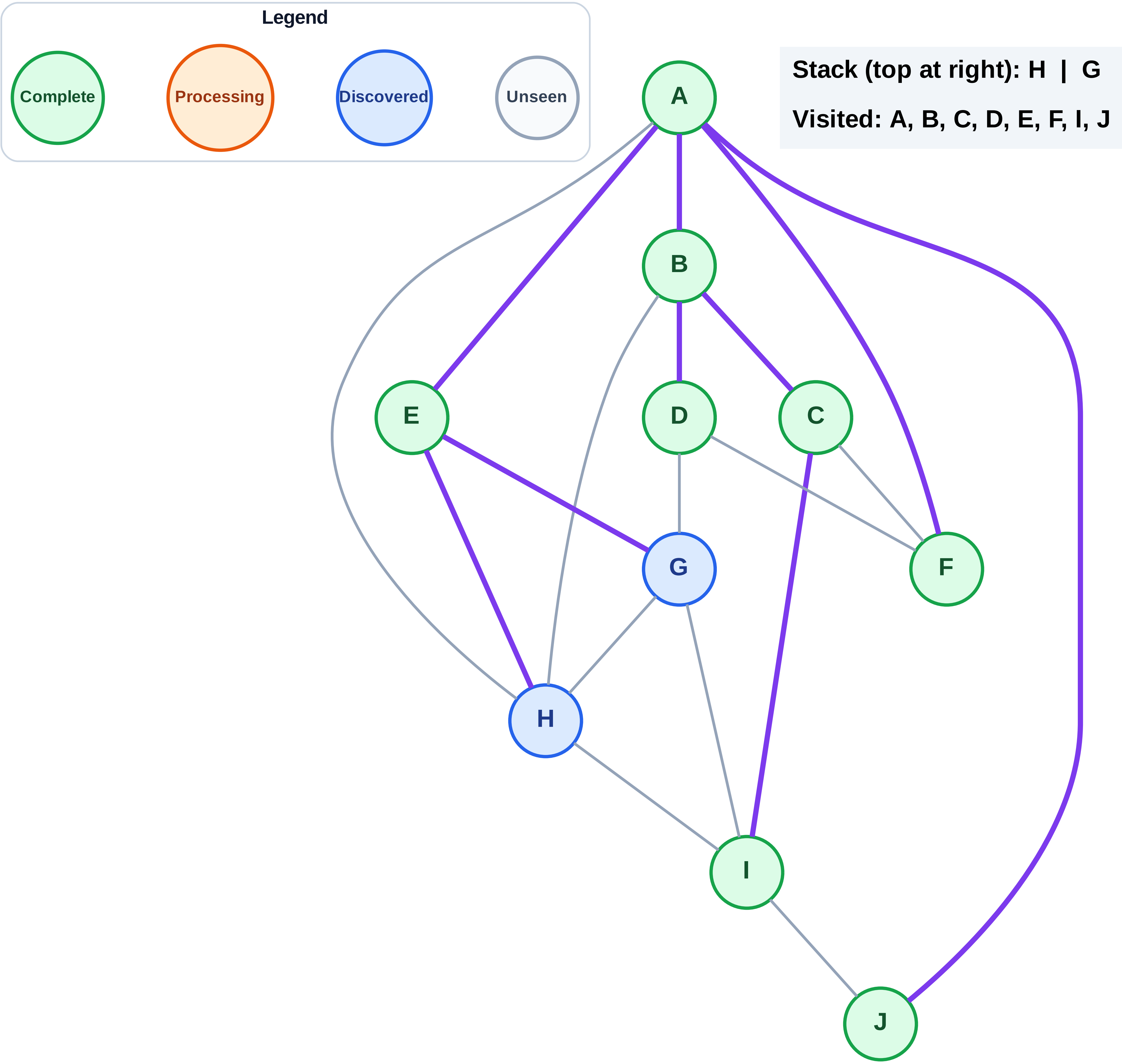
DFS Traversal

Step 16: Process node J



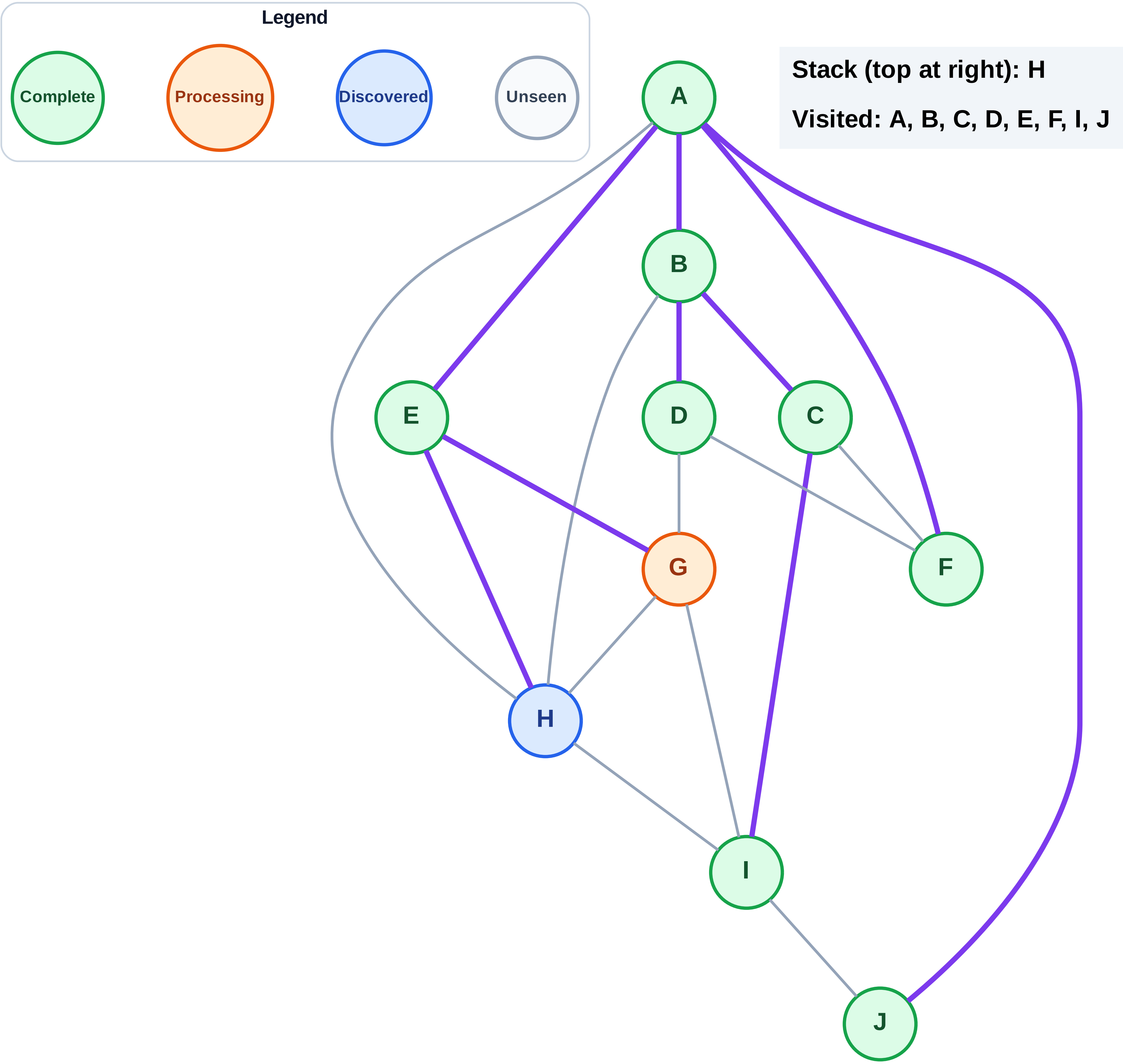
DFS Traversal

Step 17: No new neighbors from J



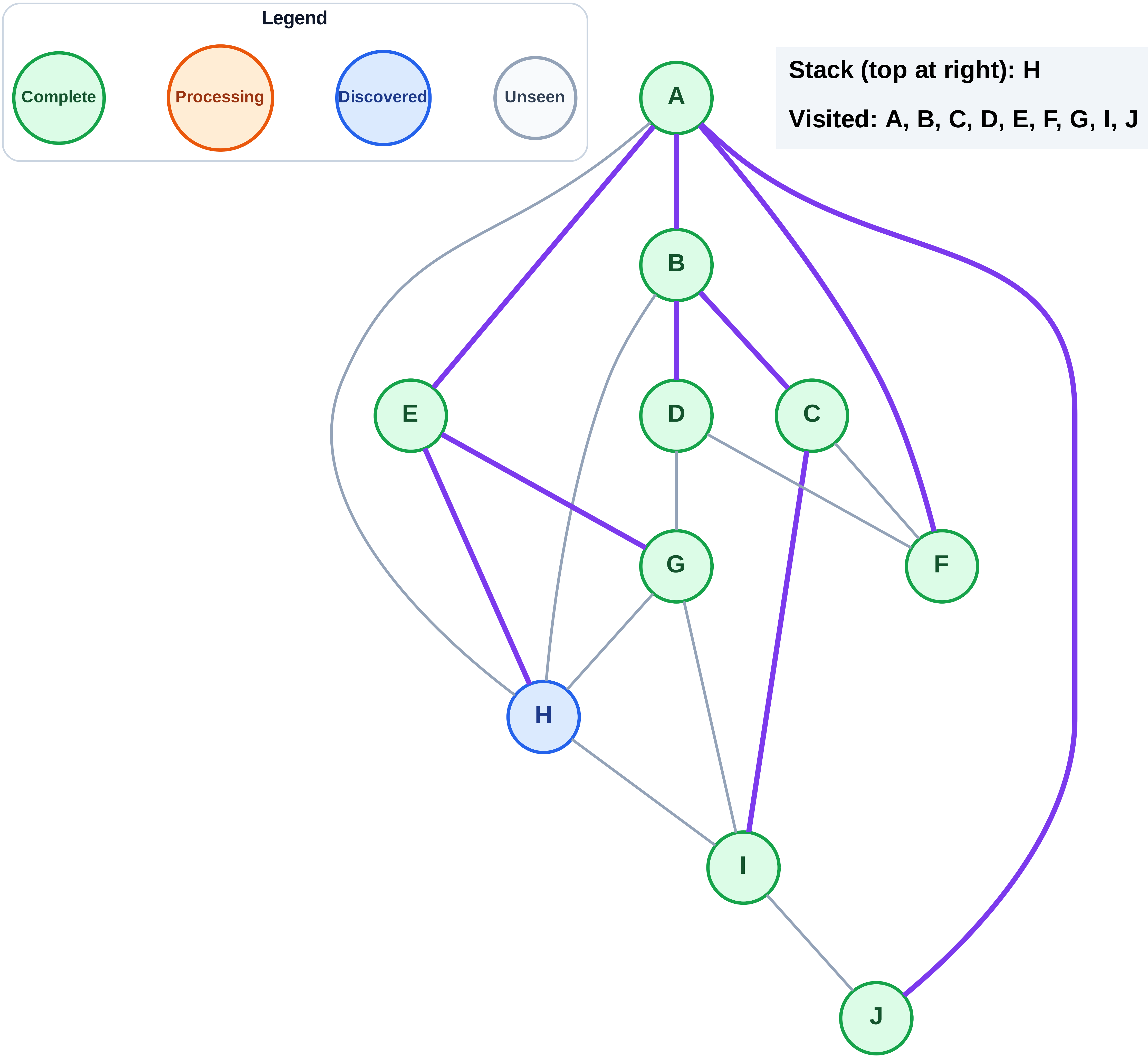
DFS Traversal

Step 18: Process node G



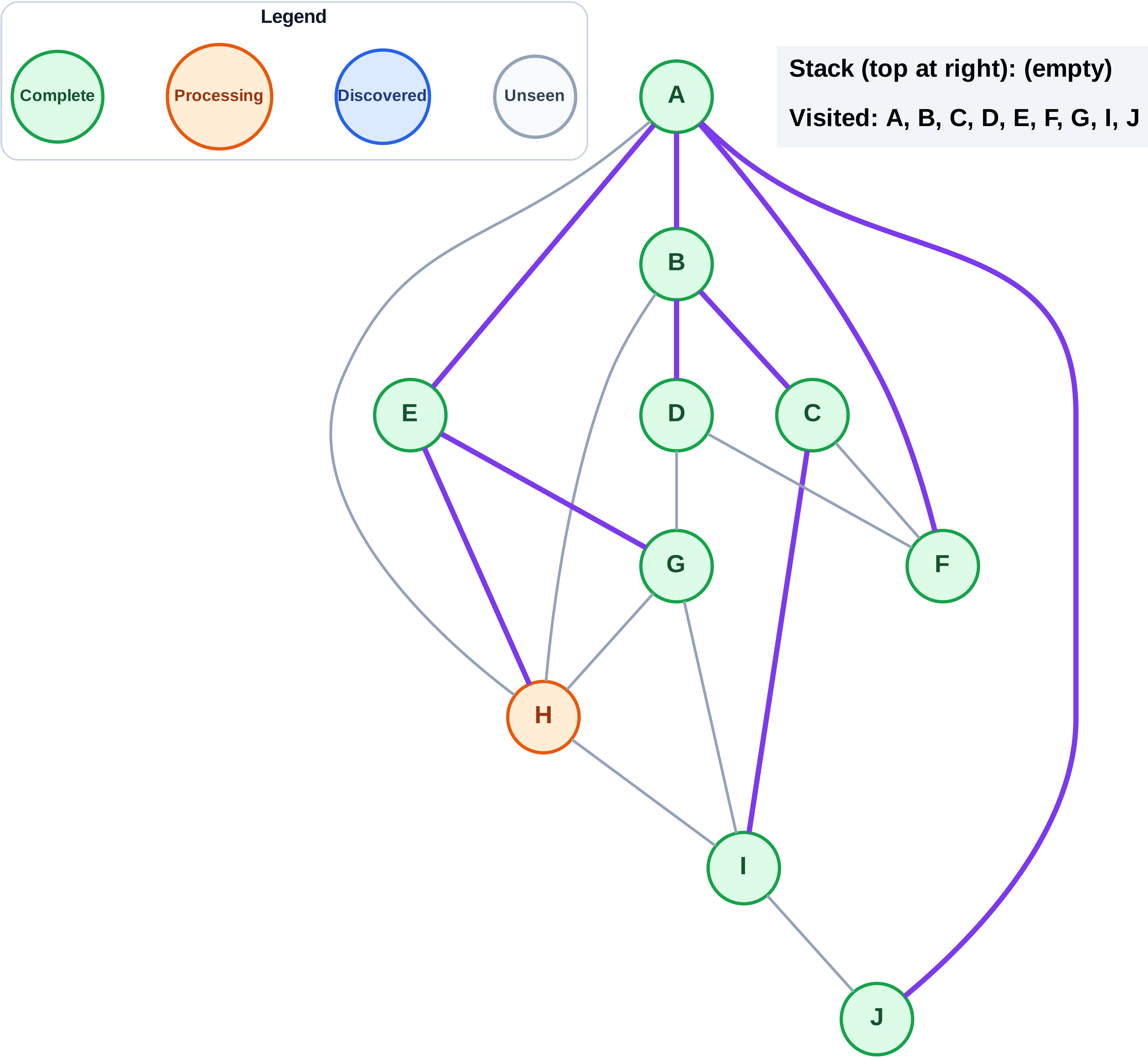
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H